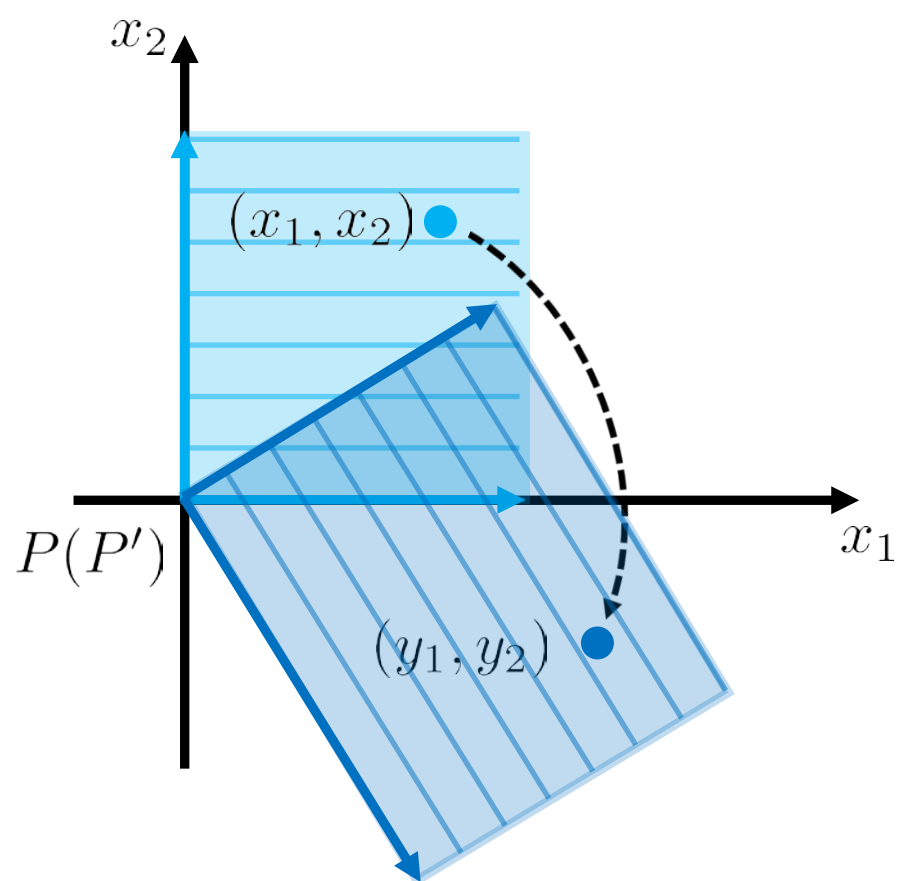
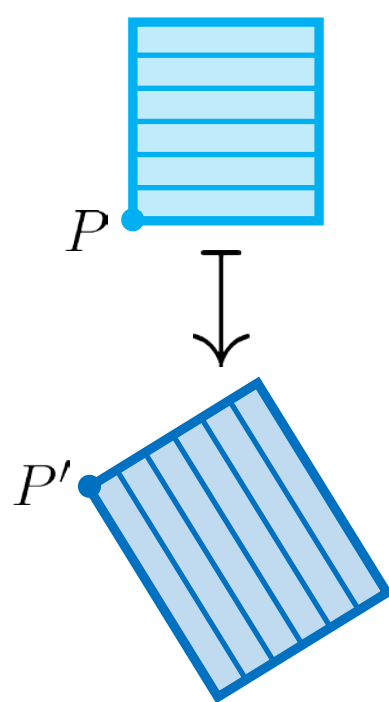
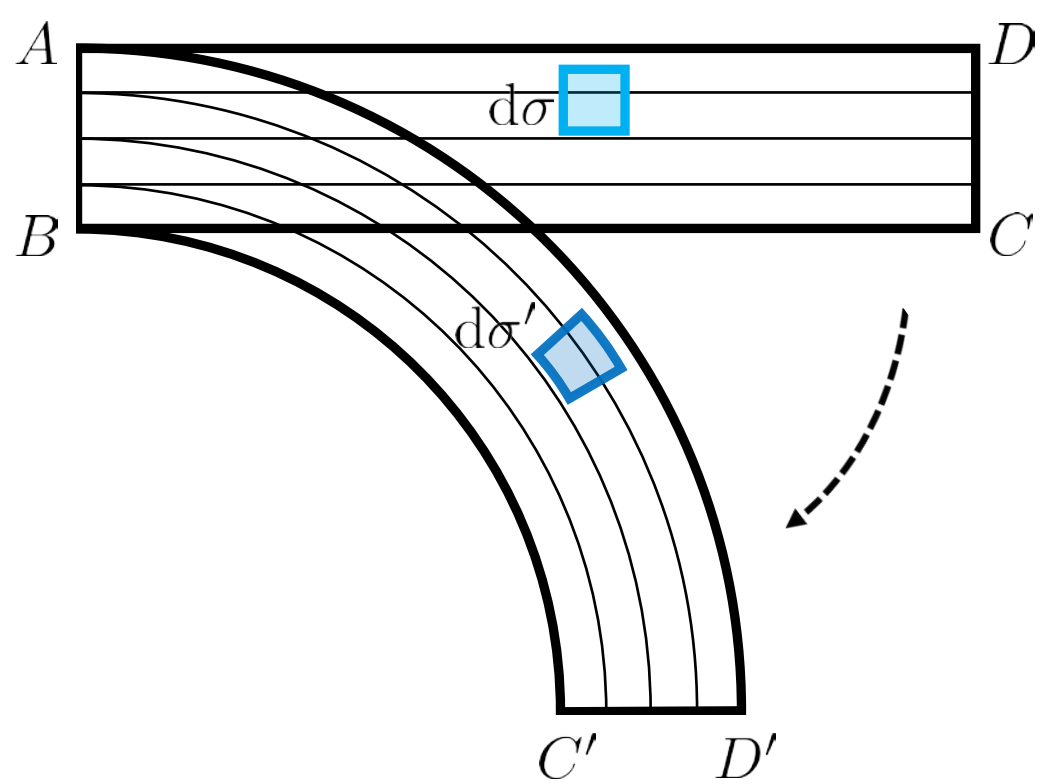
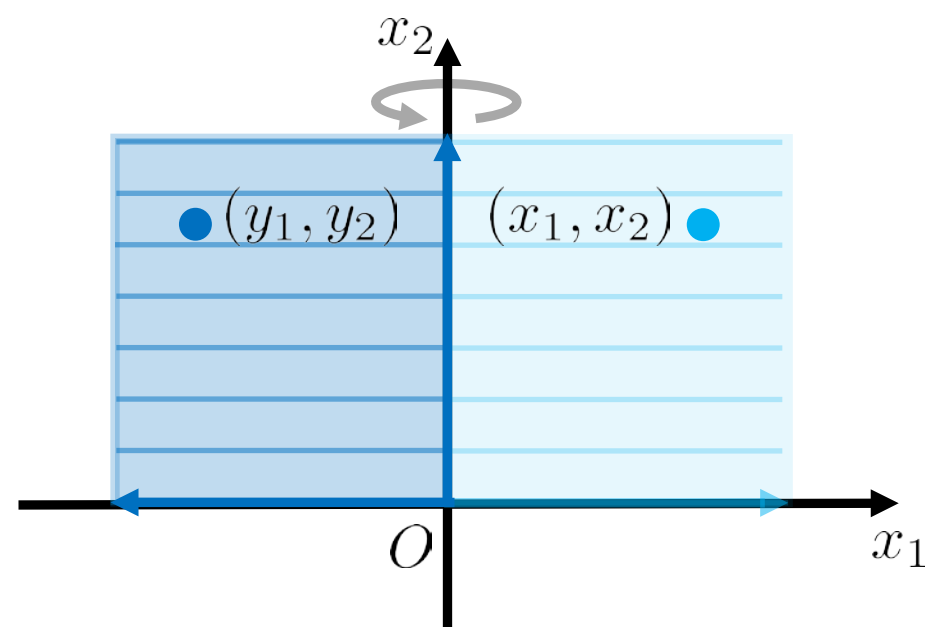
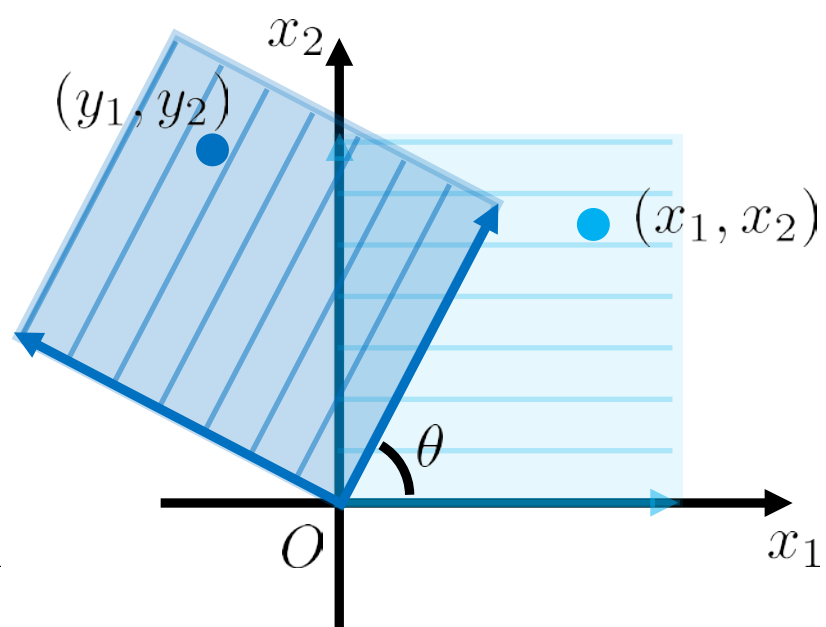
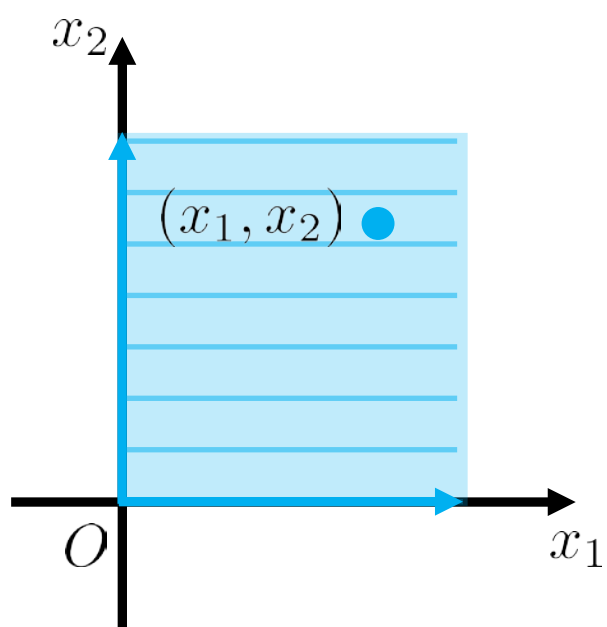
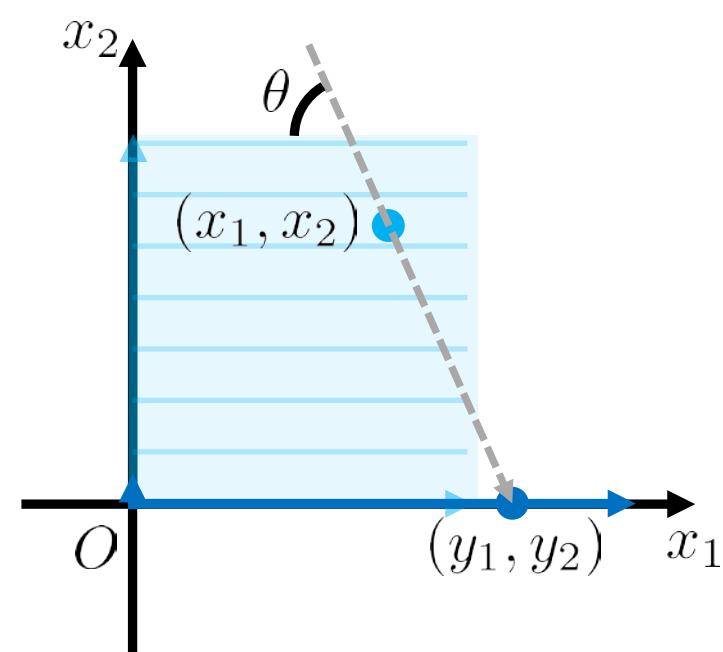
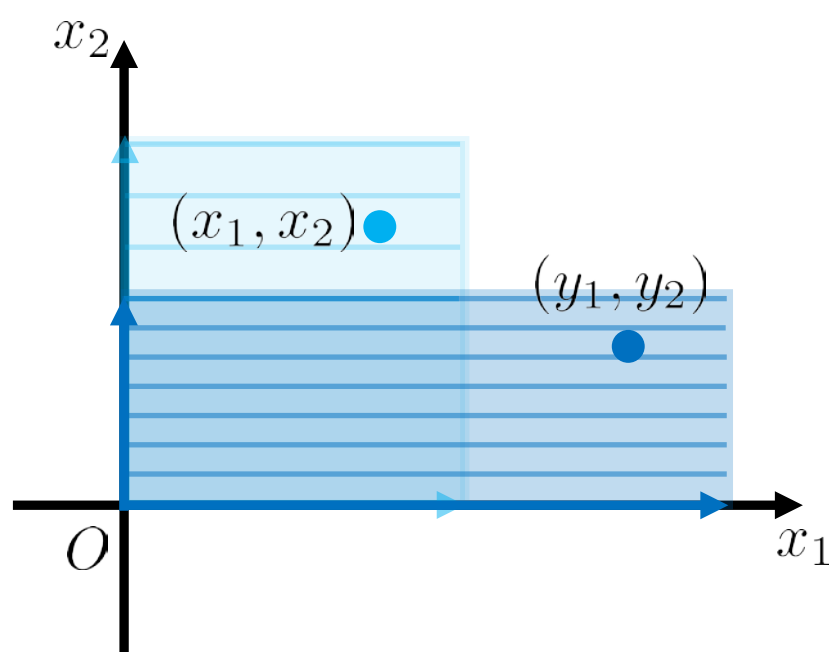
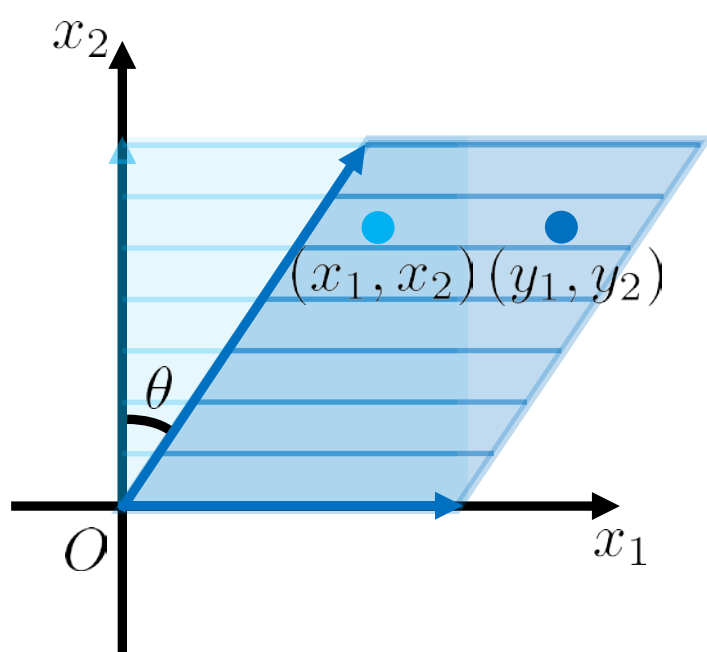






$$(1) \begin{cases} y_1 = \cos \theta \cdot x_1 - \sin \theta \cdot x_2 \\ y_2 = \sin \theta \cdot x_1 + \cos \theta \cdot x_2 \end{cases}$$

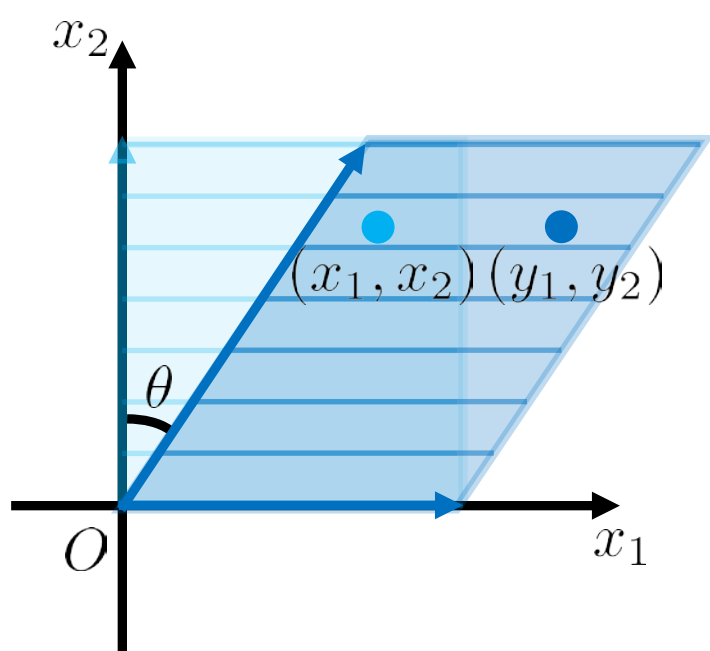
$$(2) \begin{cases} y_1 = -1 \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_1 + 1 \cdot x_2 \end{cases}$$



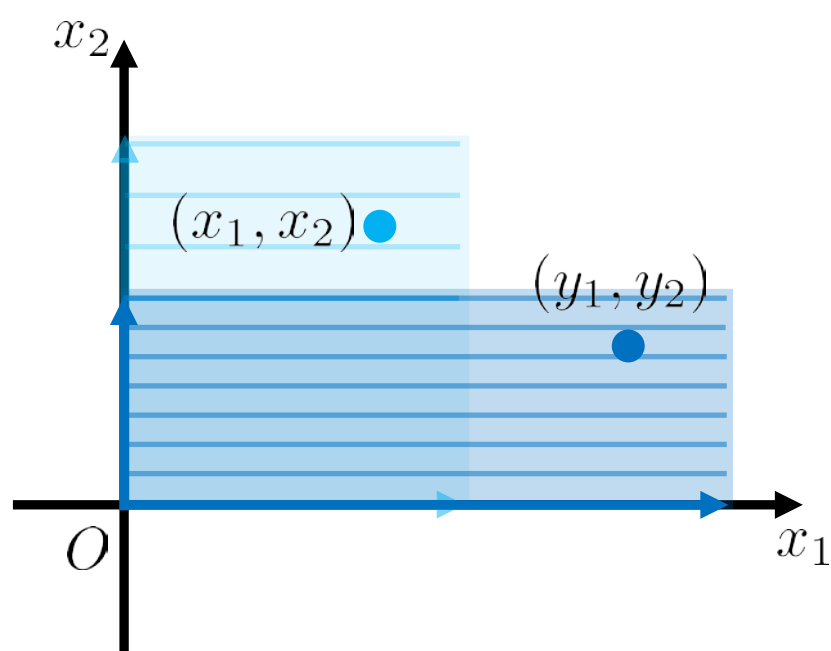
$$(3) \begin{cases} y_1 = 1 \cdot x_1 + \tan \theta \cdot x_2 \\ y_2 = 0 \cdot x_1 + 1 \cdot x_2 \end{cases}$$

$$(4) \begin{cases} y_1 = k_1 \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_1 + k_2 \cdot x_2 \end{cases}$$

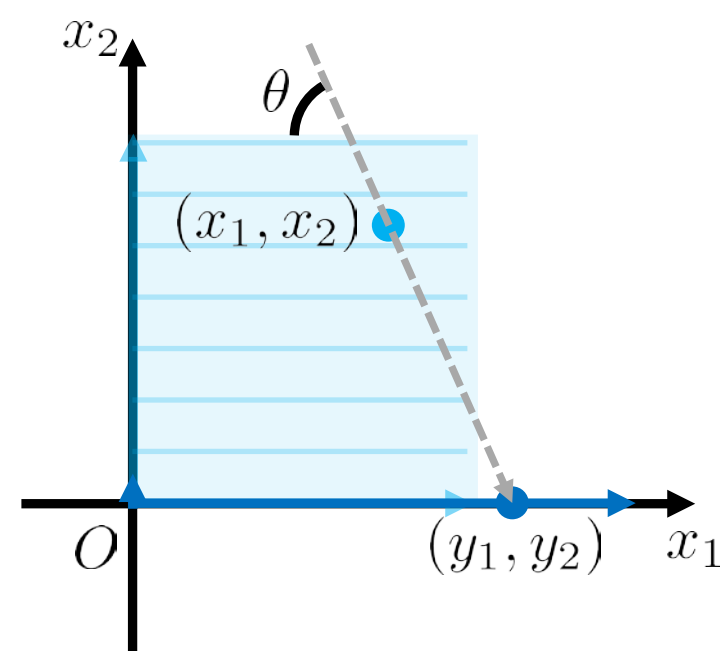
$$(5) \begin{cases} y_1 = 1 \cdot x_1 + \cot \theta \cdot x_2 \\ y_2 = 0 \cdot x_2 + 0 \cdot x_2 \end{cases}$$



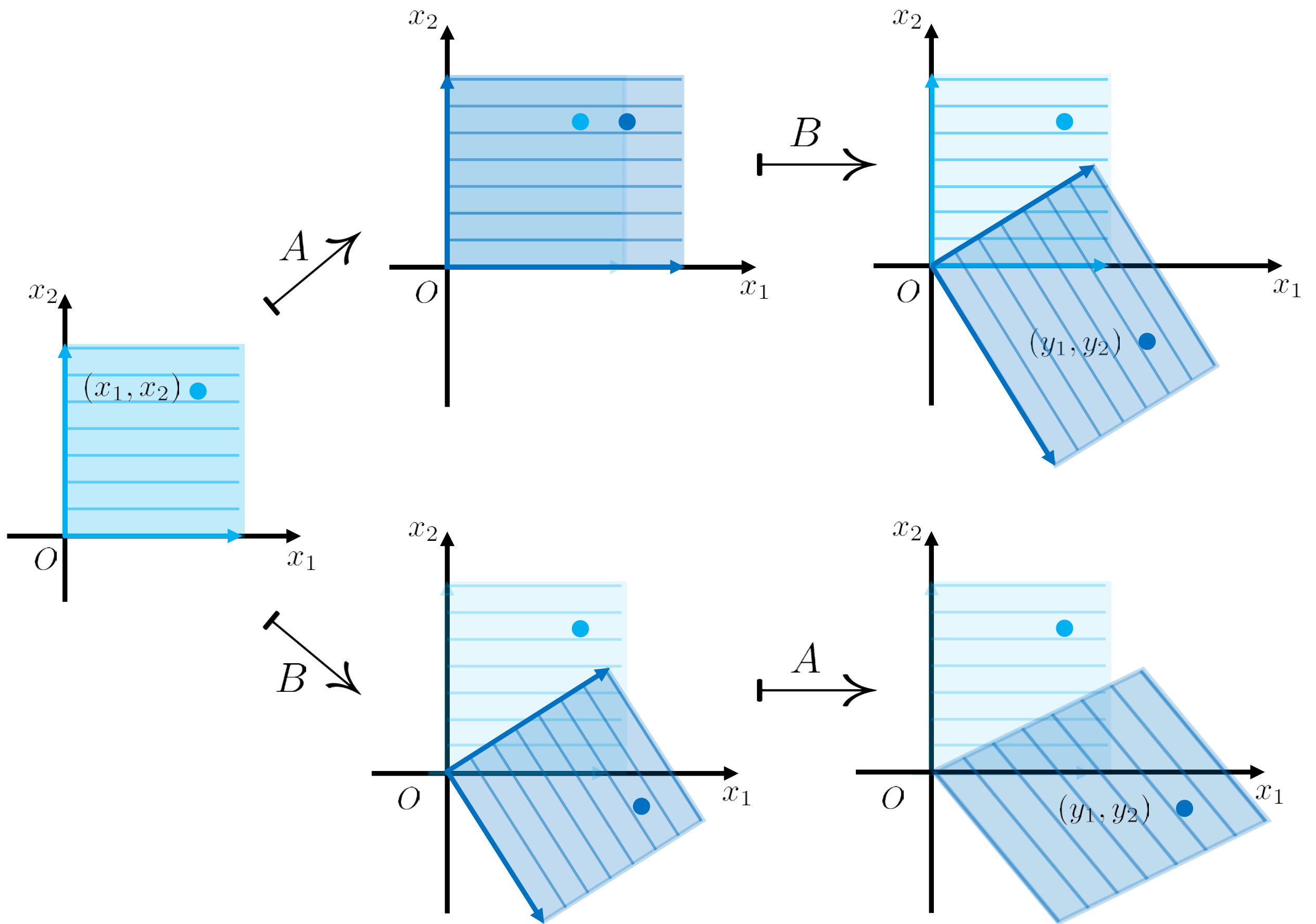
$$(3) \begin{cases} y_1 = 1 \cdot x_1 + \tan \theta \cdot x_2 \\ y_2 = 0 \cdot x_1 + 1 \cdot x_2 \end{cases}$$

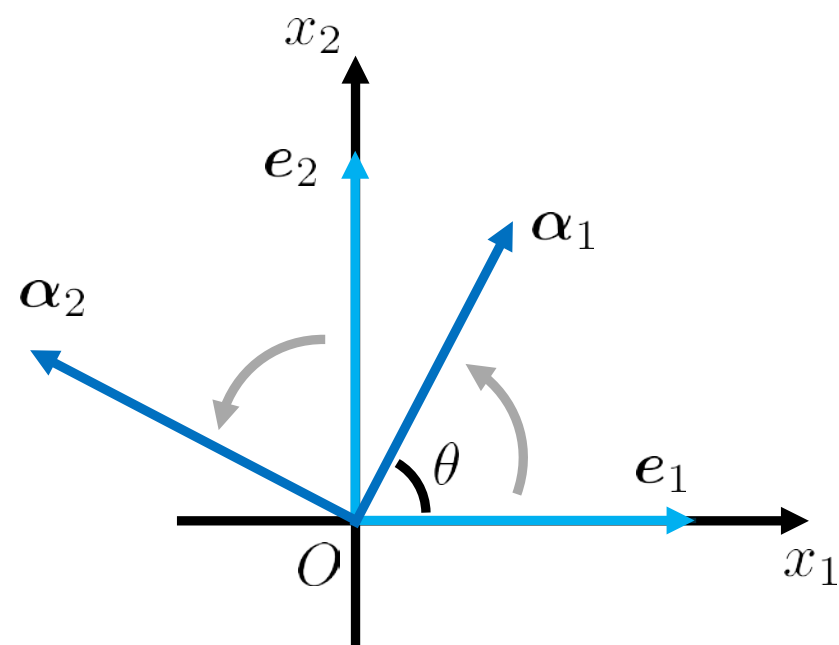
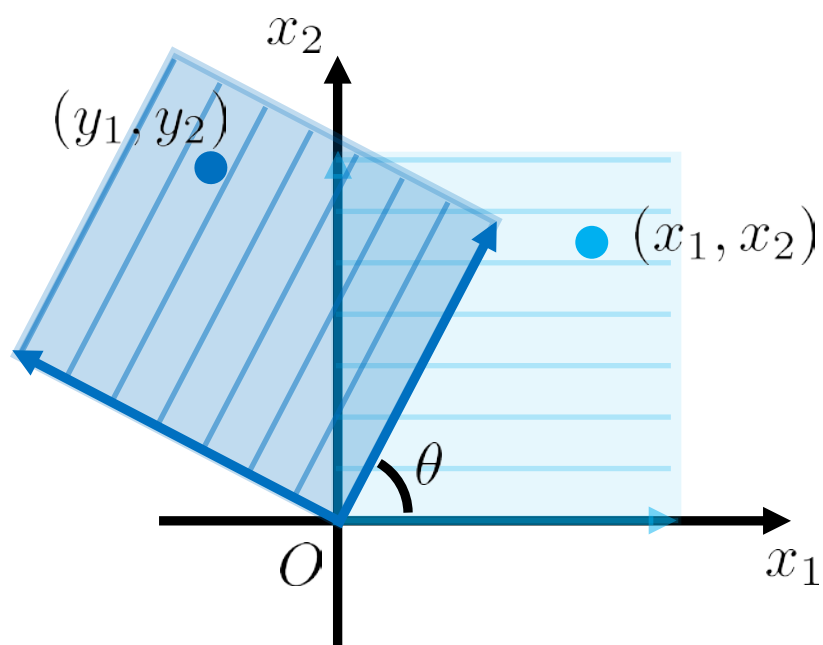


$$(4) \begin{cases} y_1 = k_1 \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_1 + k_2 \cdot x_2 \end{cases}$$



$$(5) \begin{cases} y_1 = (1 + \cot \theta) \cdot x_1 + 0 \cdot x_2 \\ y_2 = 0 \cdot x_1 + 0 \cdot x_2 \end{cases}$$

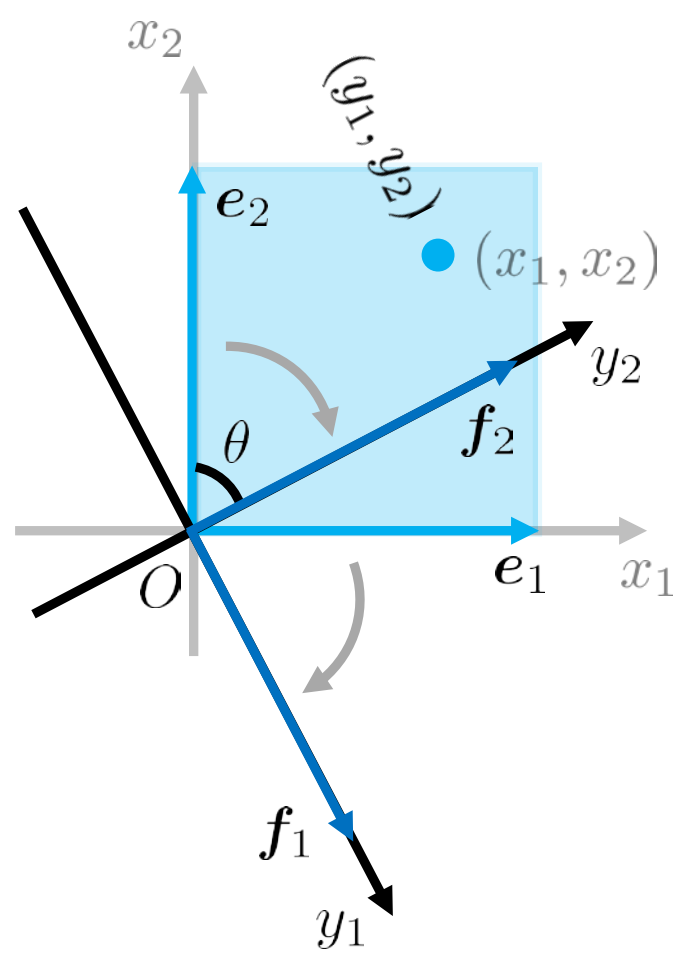
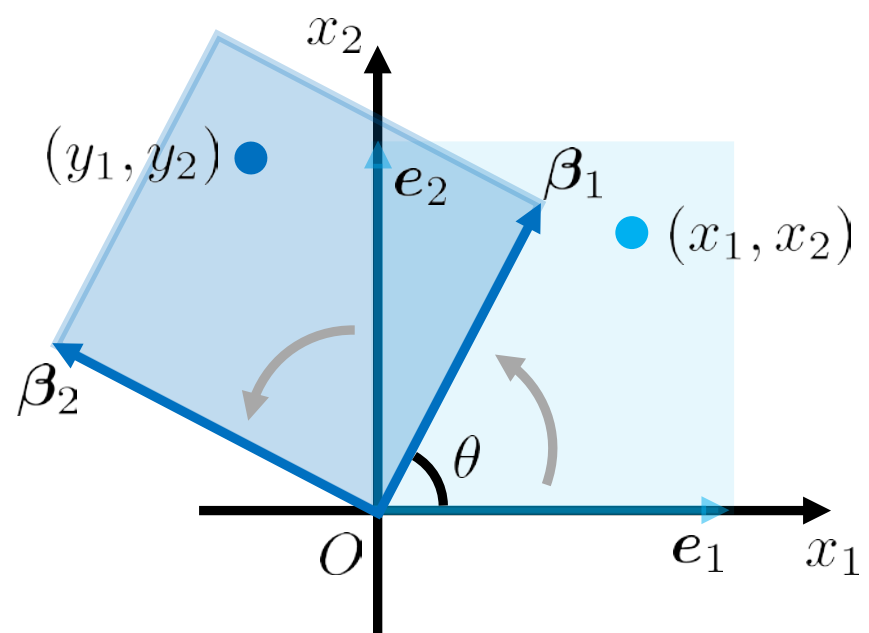


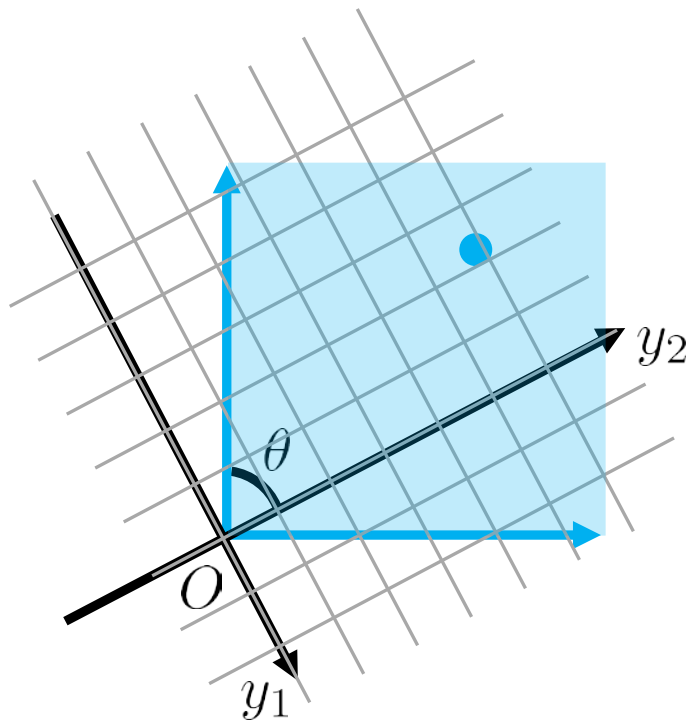
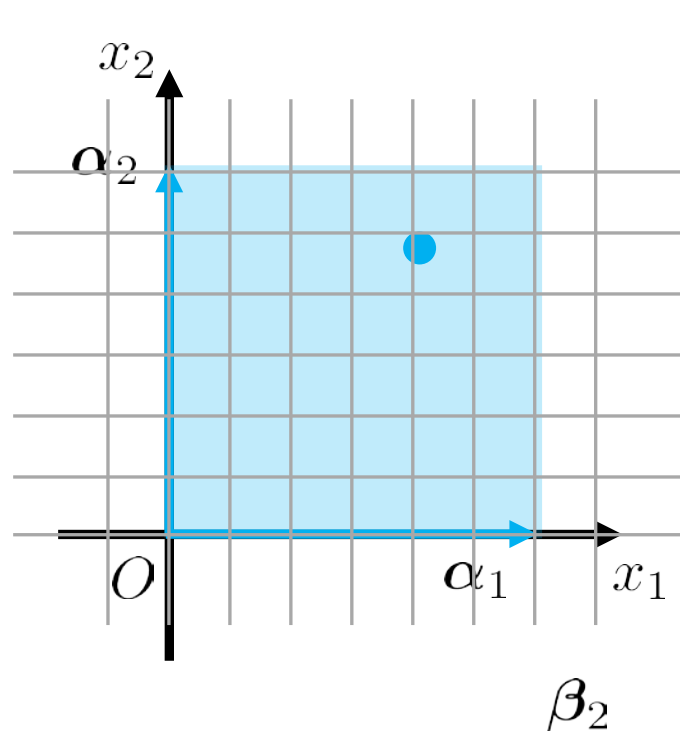


$$\begin{cases} y_1 = \cos \theta \cdot x_1 - \sin \theta \cdot x_2 \\ y_2 = \sin \theta \cdot x_1 + \cos \theta \cdot x_2 \end{cases} \iff \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

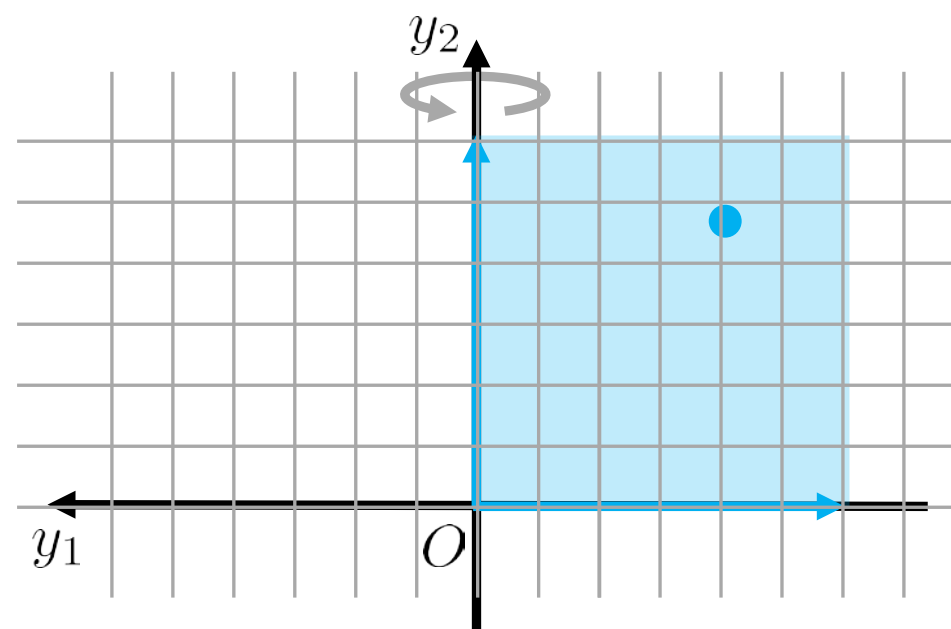
$$\iff \mathbf{y} = A\mathbf{x} = (\boldsymbol{\alpha}_1 \ \boldsymbol{\alpha}_2)\mathbf{x}$$



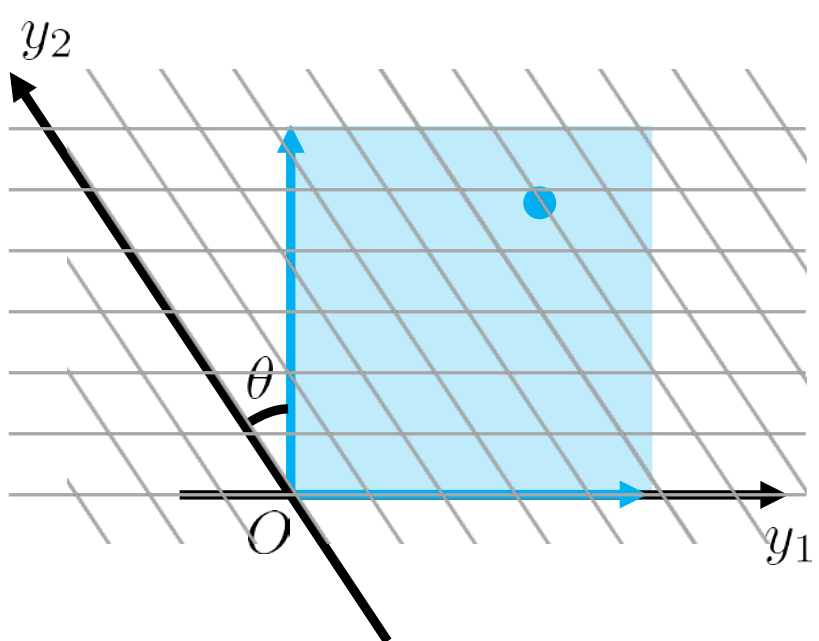




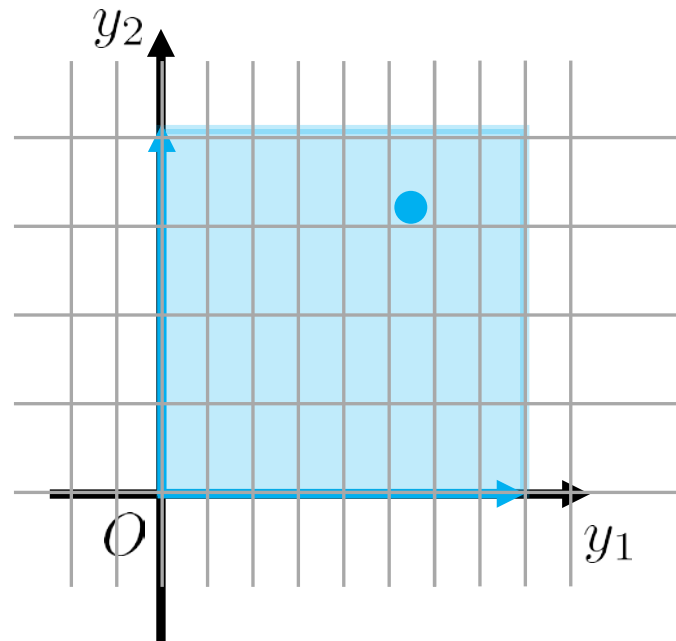
$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



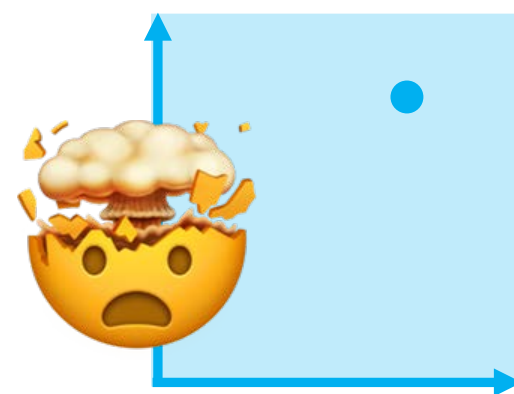
$$F = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$



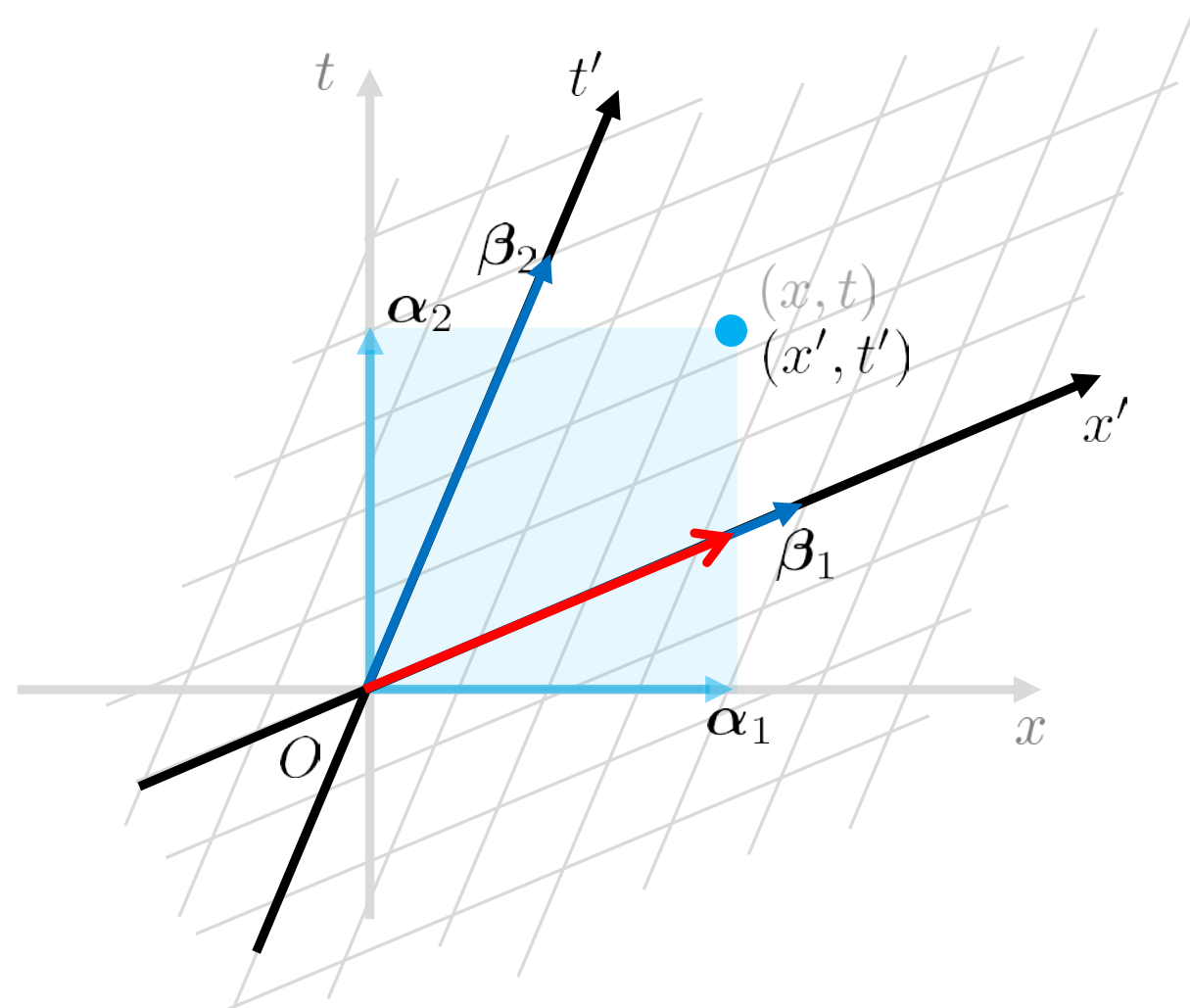
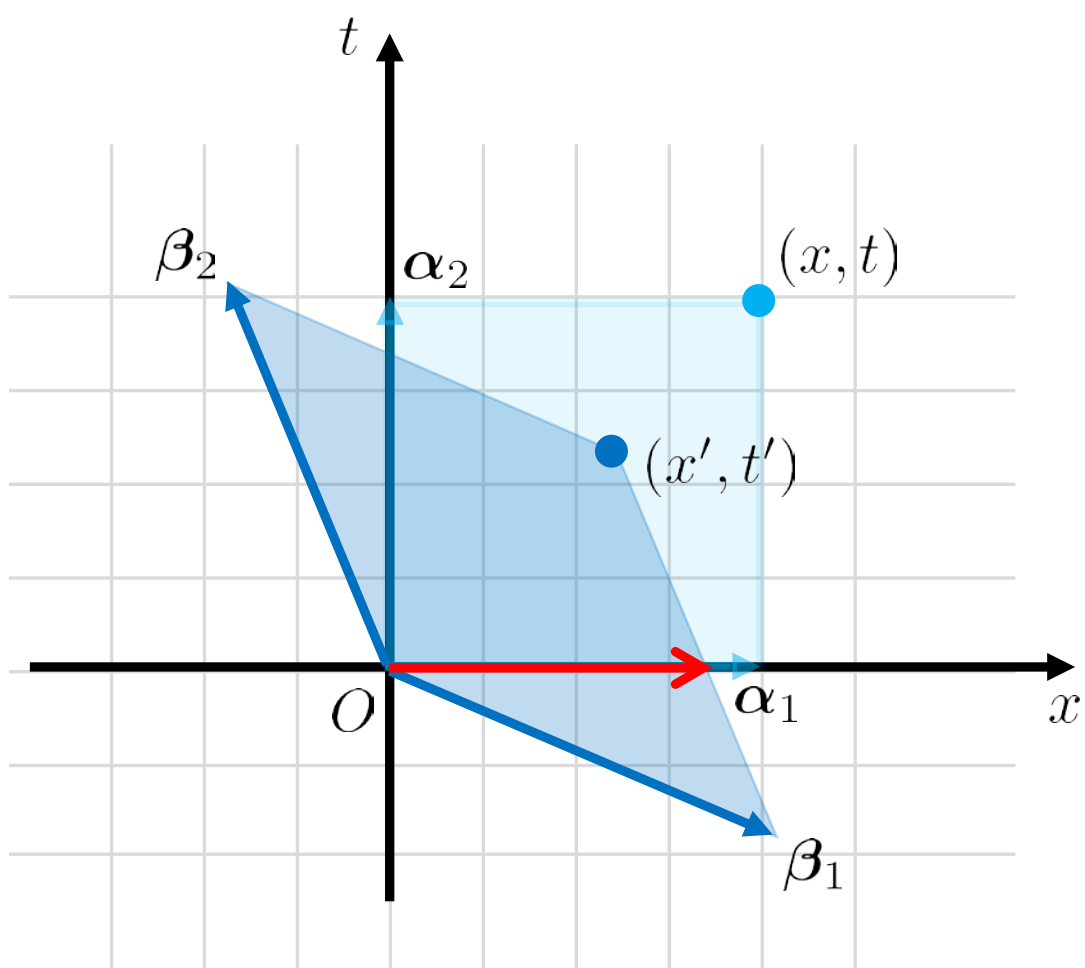
$$S = \begin{pmatrix} 1 & \tan \theta \\ 0 & 1 \end{pmatrix}$$



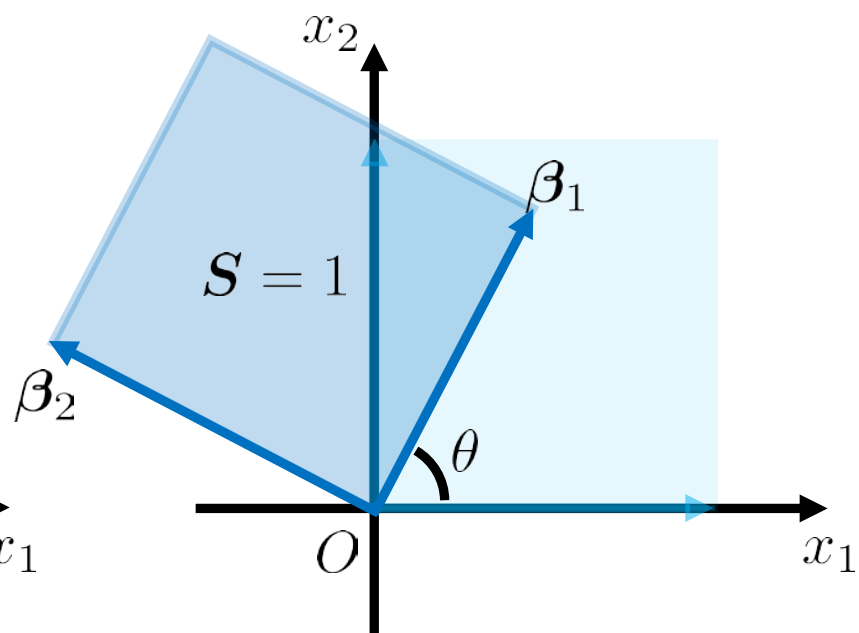
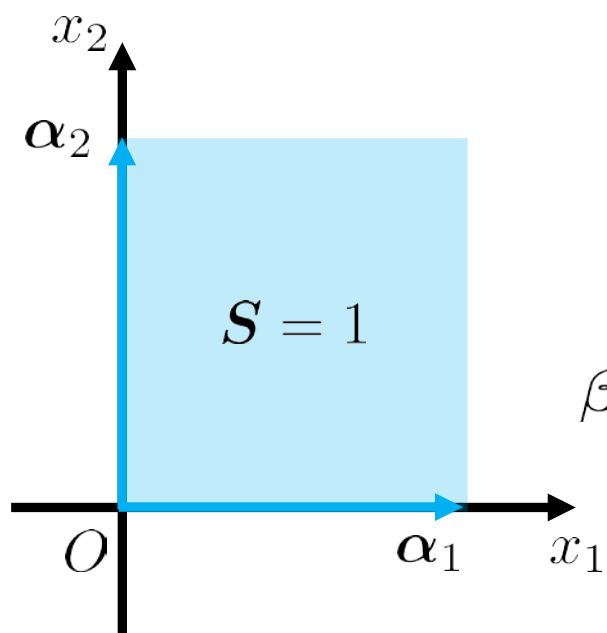
$$D = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix}$$



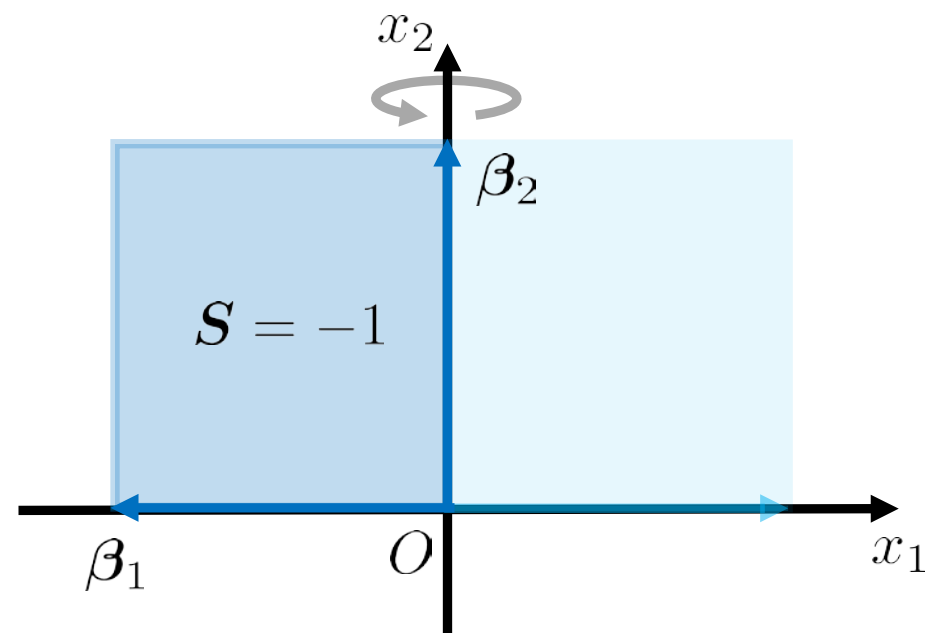
$$P = \begin{pmatrix} 1 & \cot \theta \\ 0 & 0 \end{pmatrix}$$



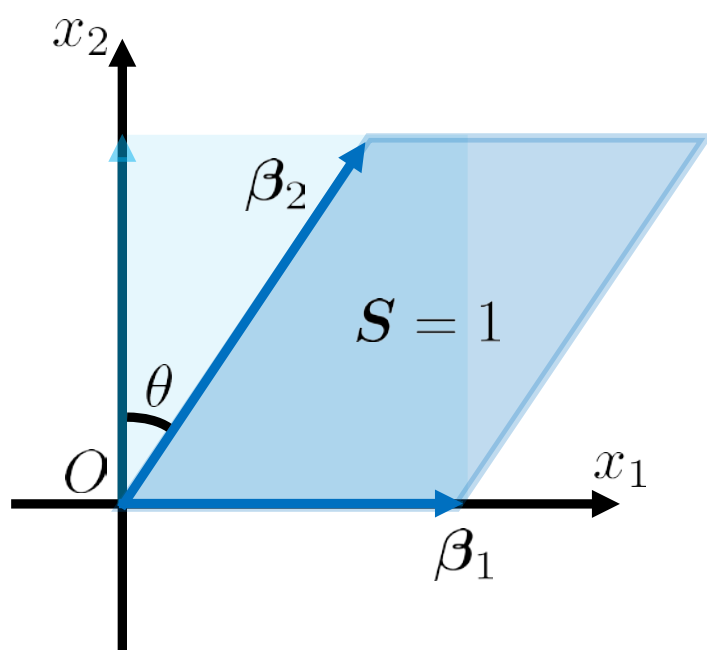




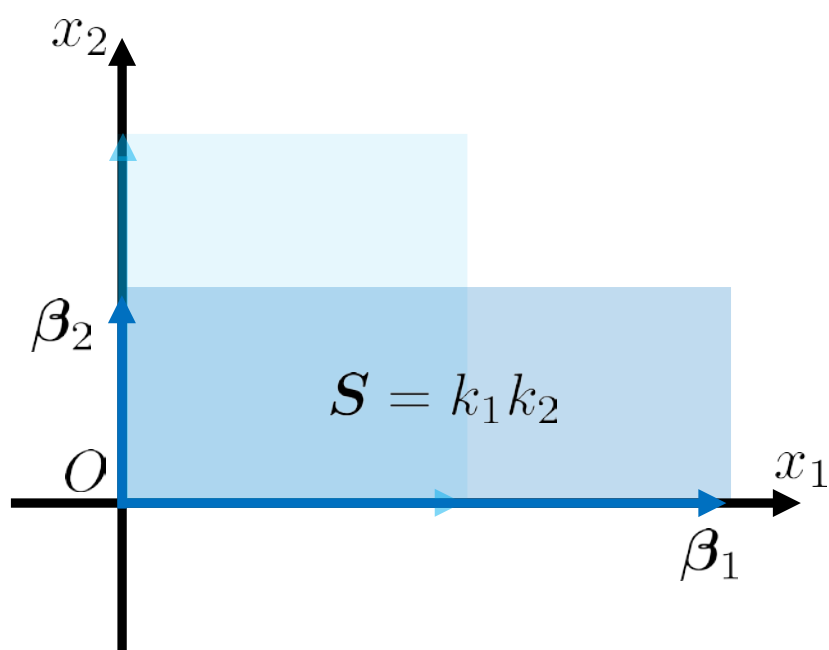
$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$



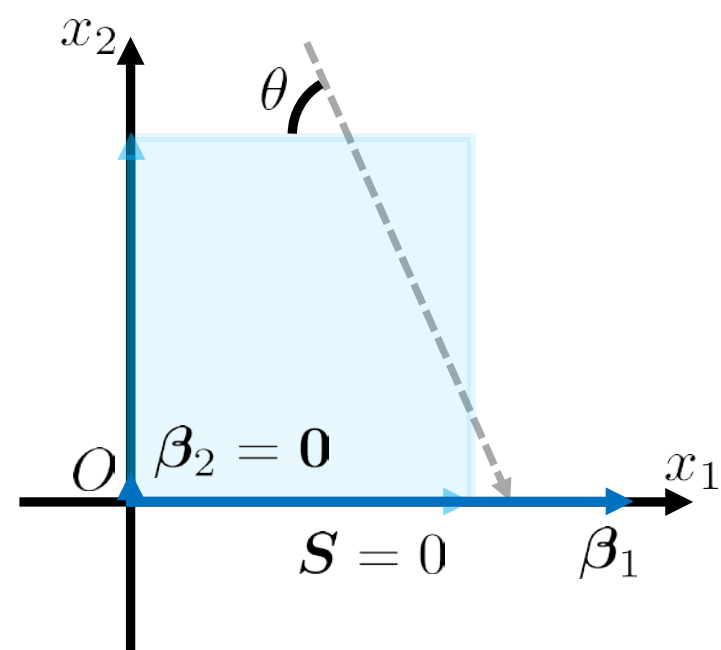
$$F = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$



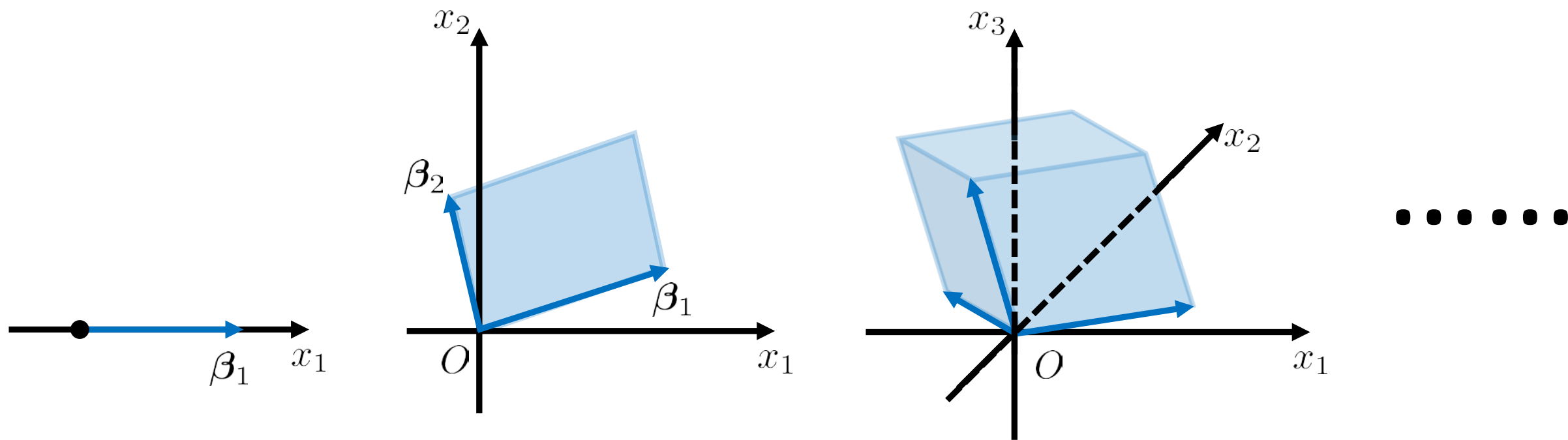
$$S = \begin{pmatrix} 1 & \tan \theta \\ 0 & 1 \end{pmatrix}$$

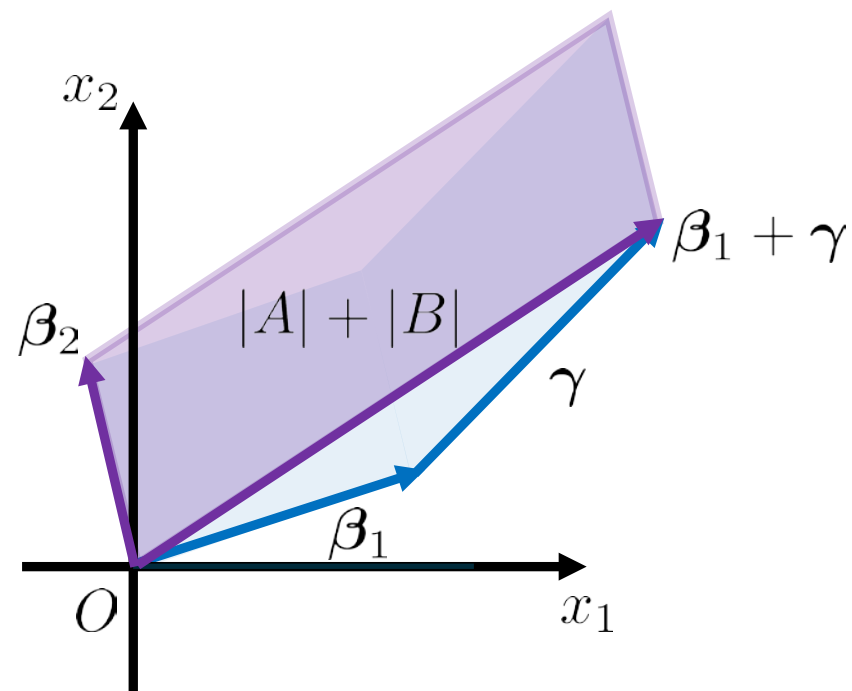
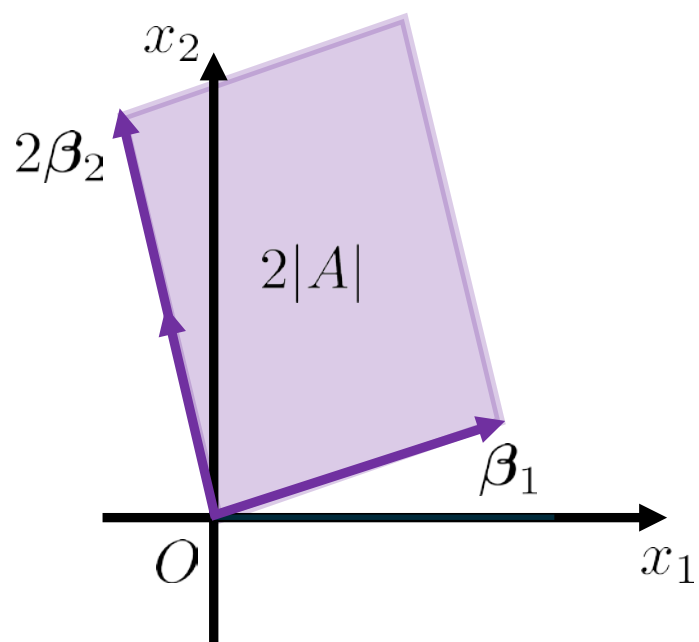
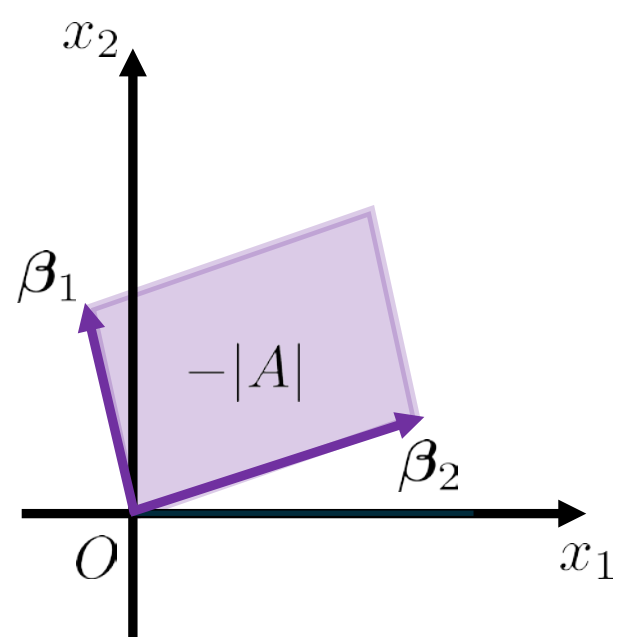
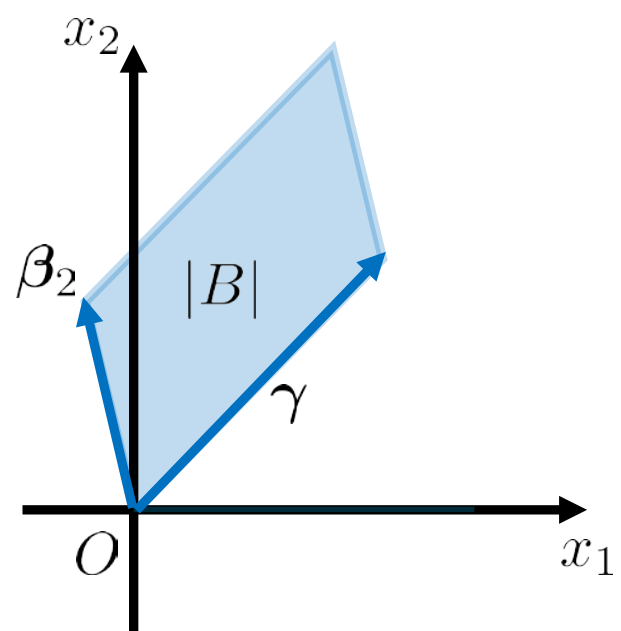
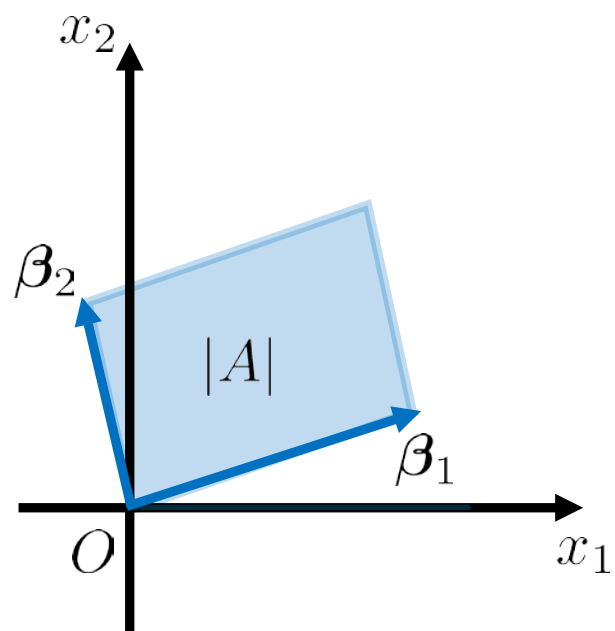
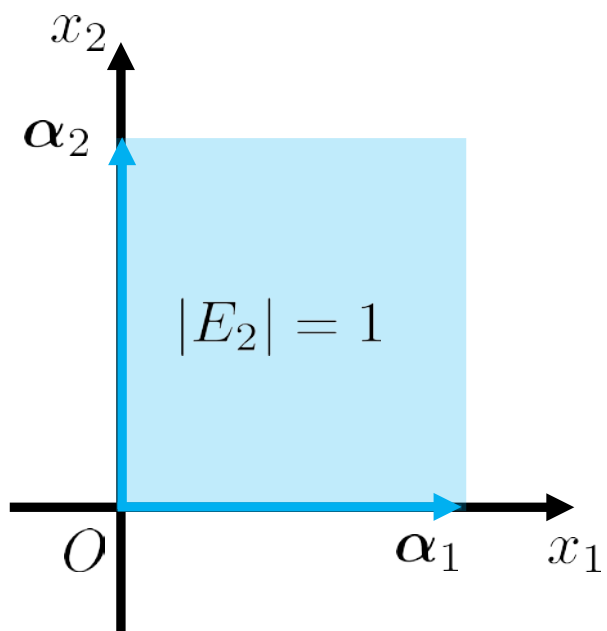


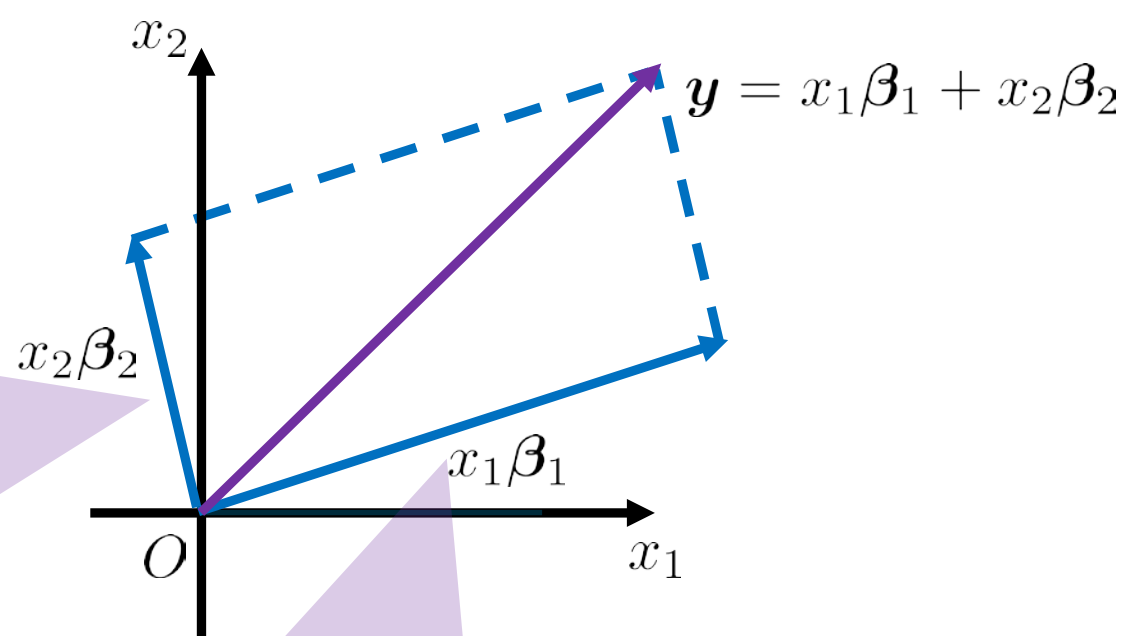
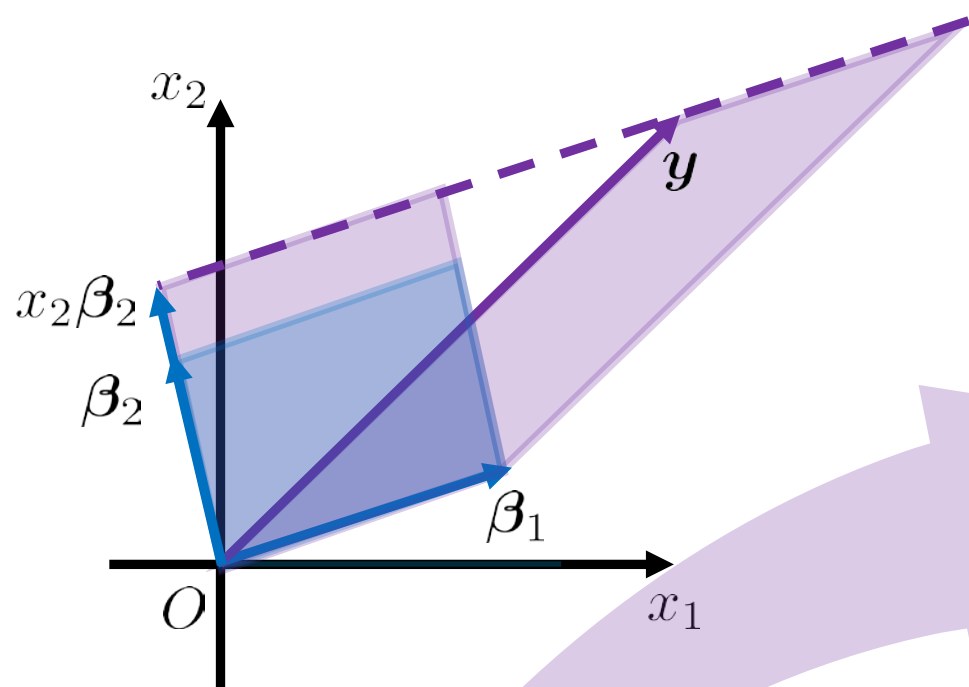
$$D = \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix}$$



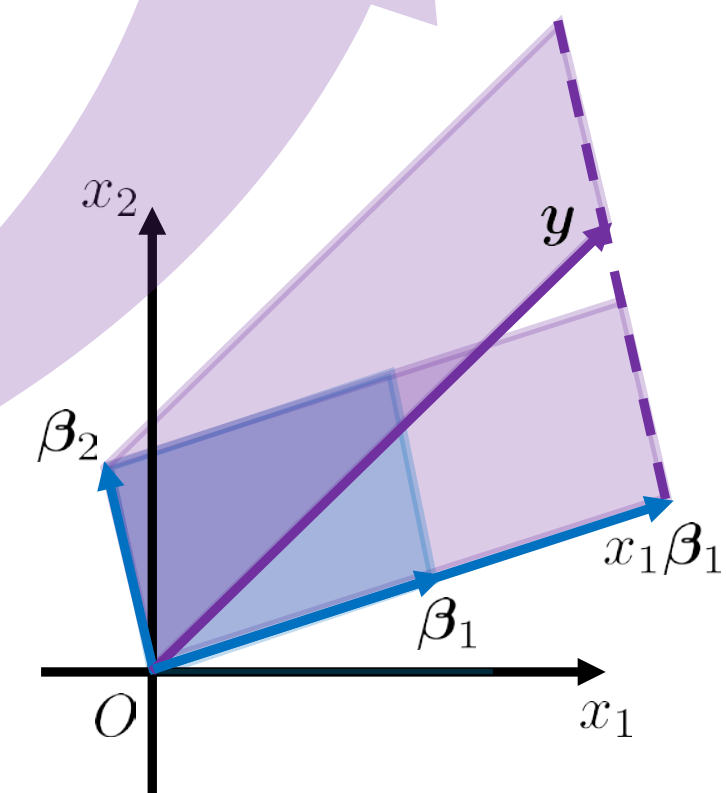
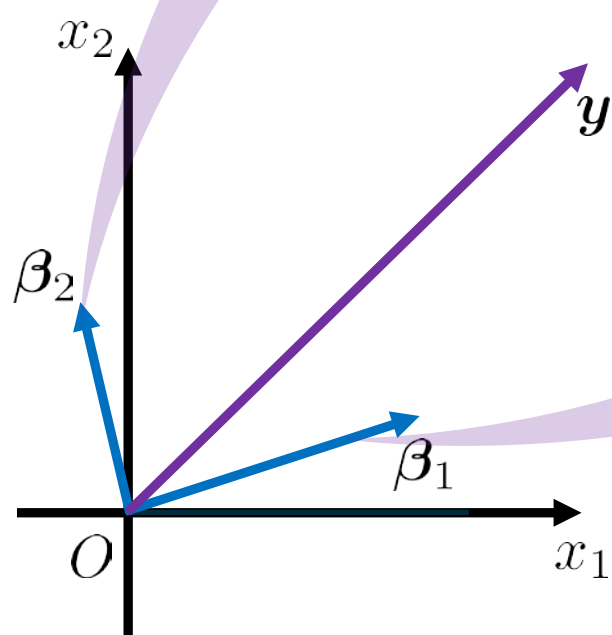
$$P = \begin{pmatrix} 1 & \cot \theta \\ 0 & 0 \end{pmatrix}$$



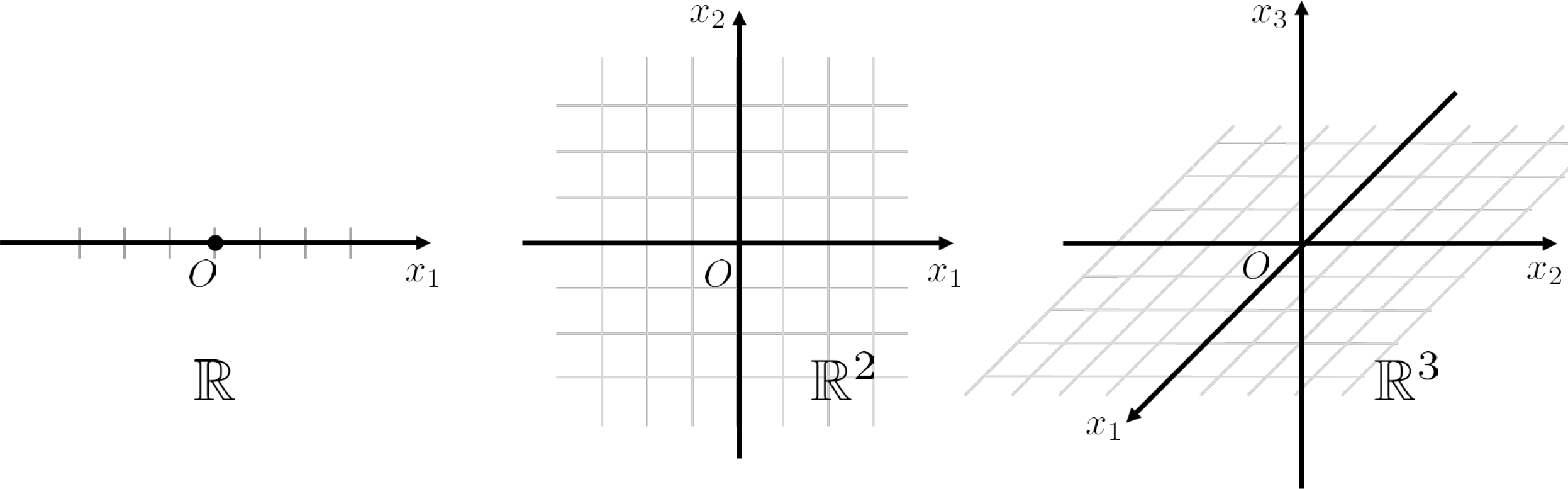




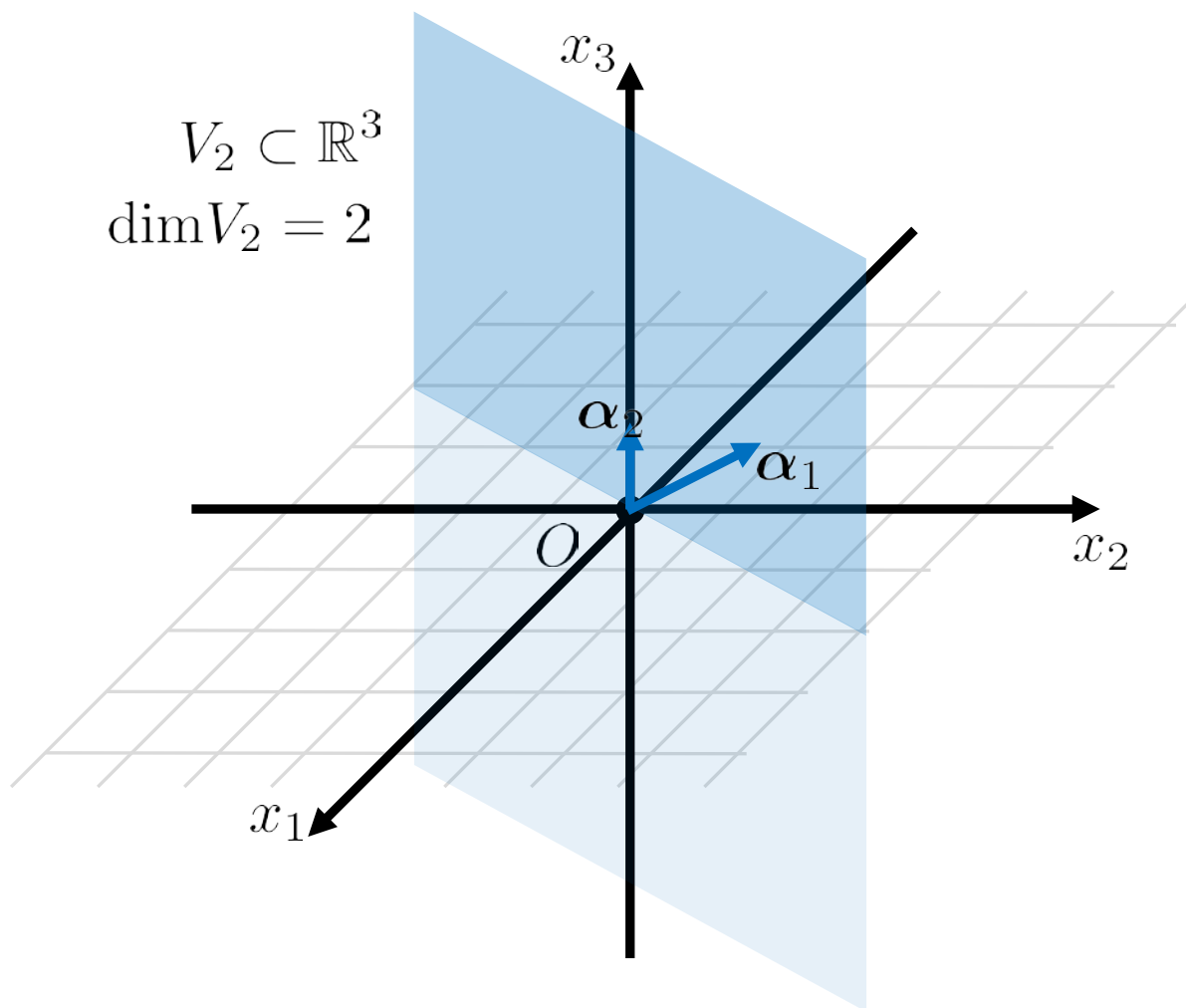
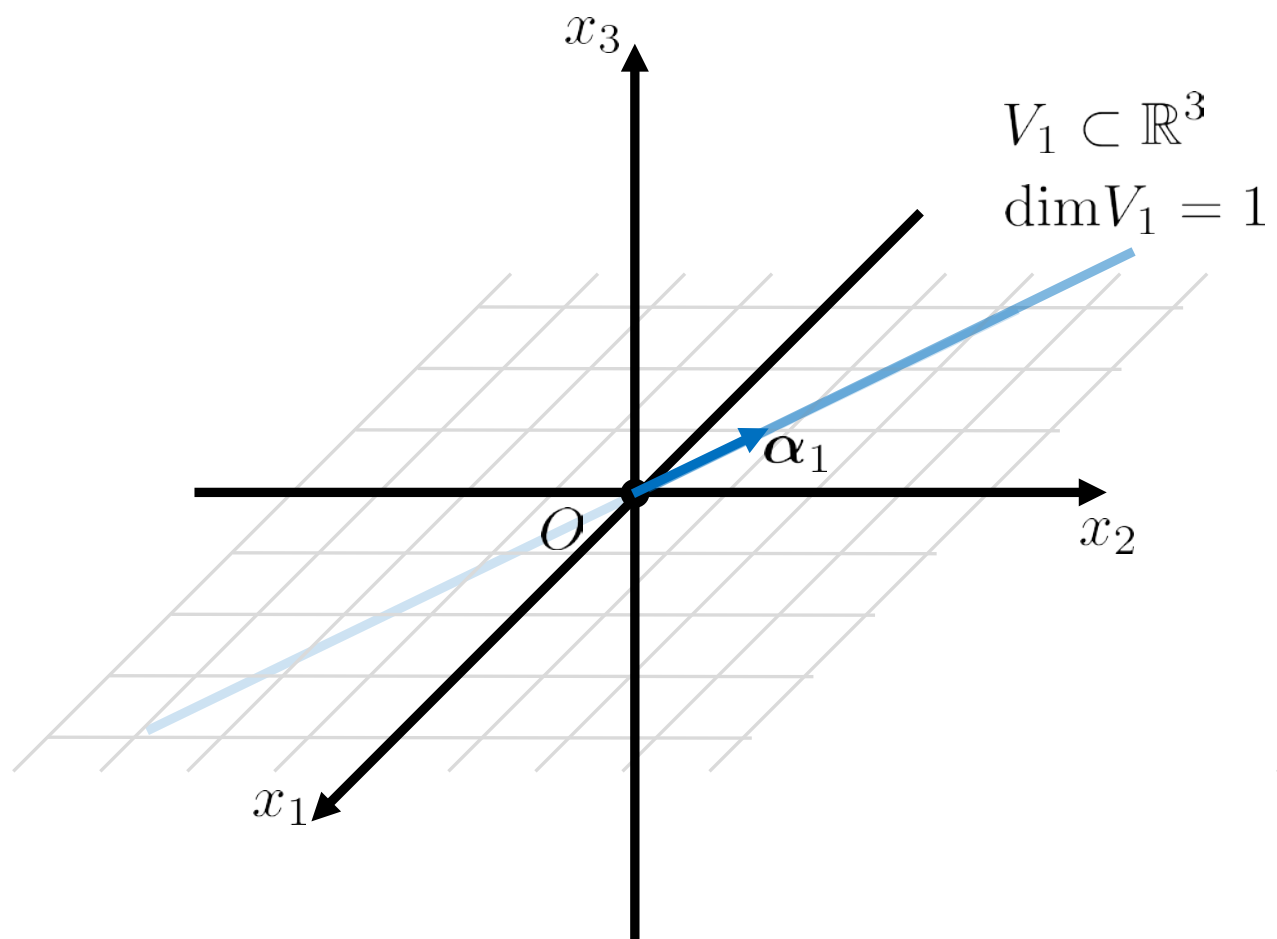
$$x_2 = \frac{|x_2\beta_2|}{|\beta_2|} = \frac{\det(\beta_1, y)}{\det(\beta_1, \beta_2)}$$

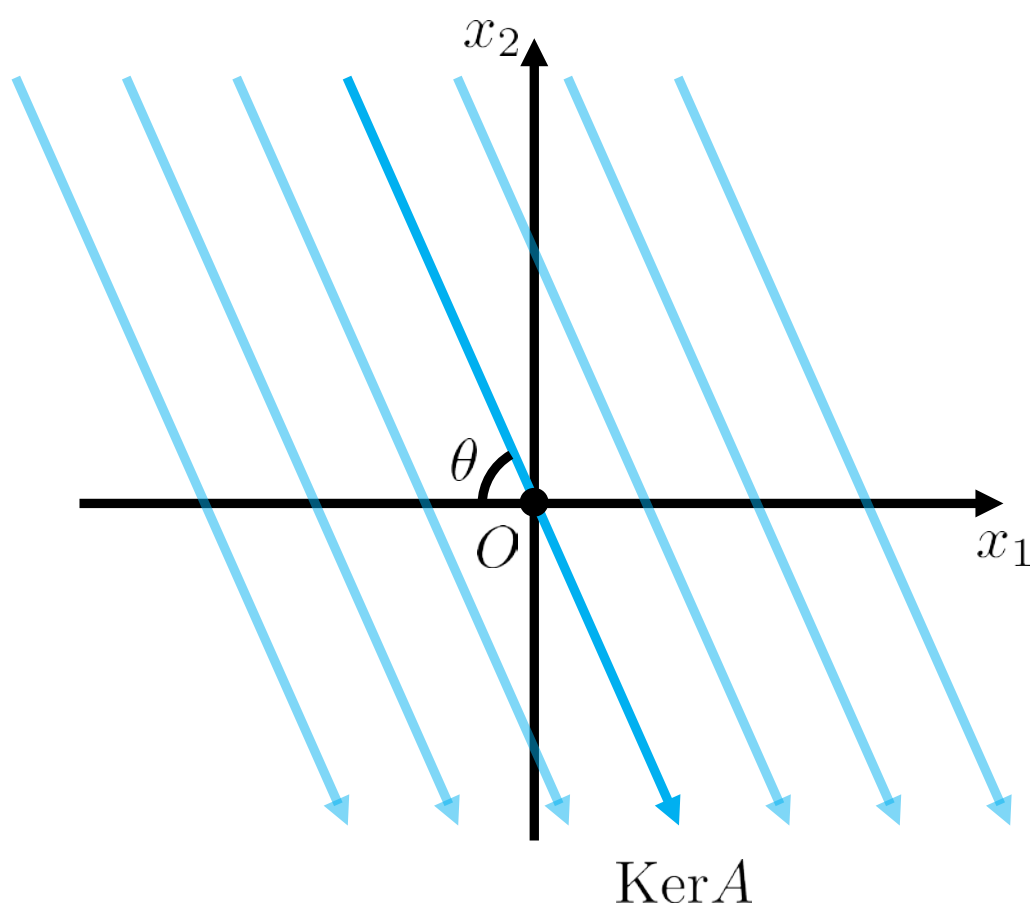


$$x_1 = \frac{|x_1\beta_1|}{|\beta_1|} = \frac{\det(y, \beta_2)}{\det(\beta_1, \beta_2)}$$

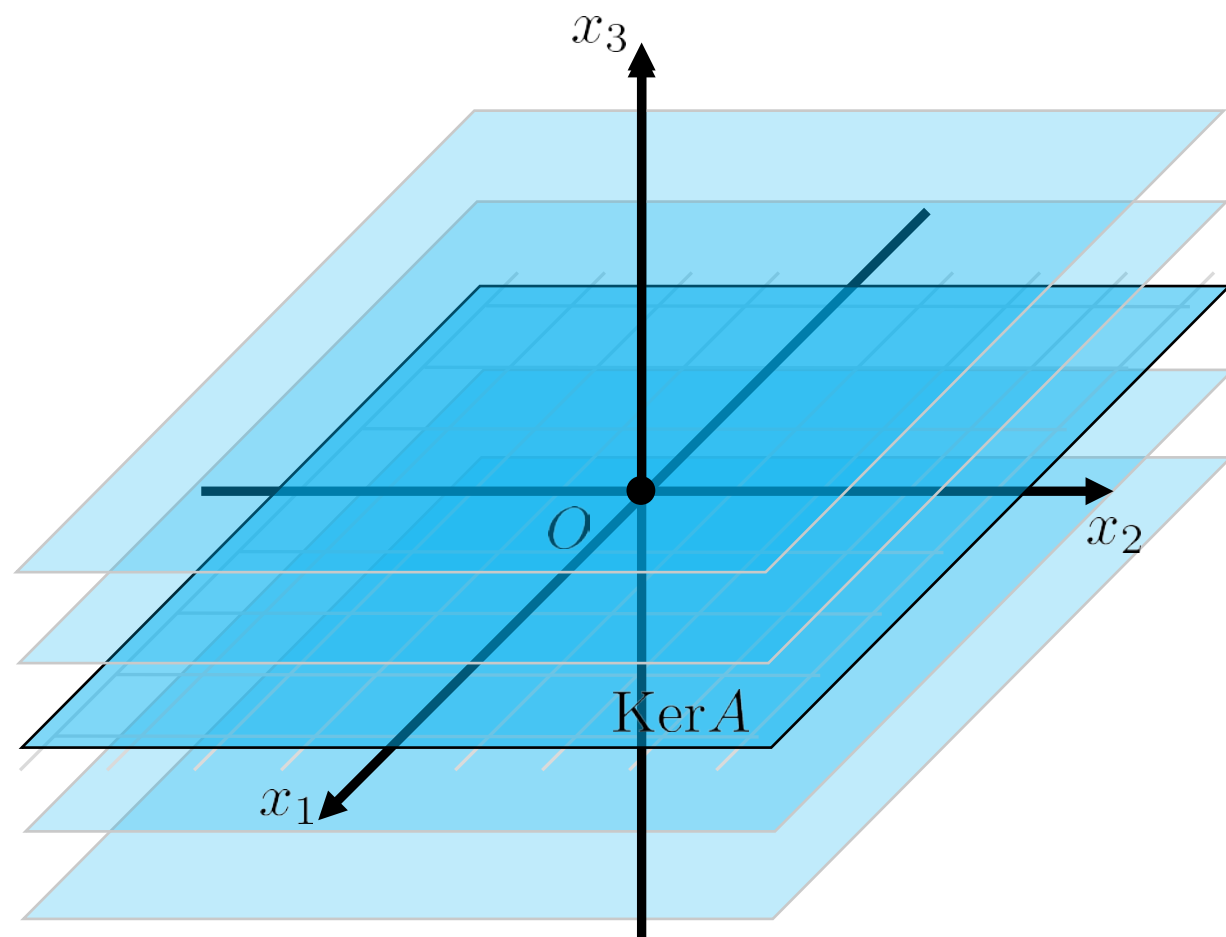




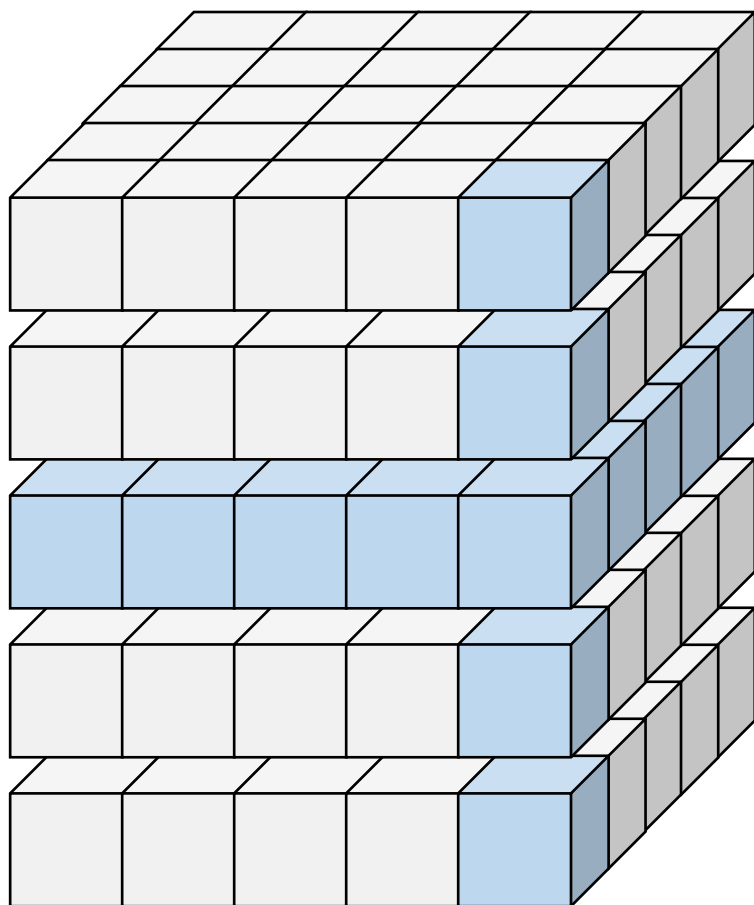




$$A = \begin{pmatrix} 1 & \cot \theta \\ 0 & 0 \end{pmatrix}$$

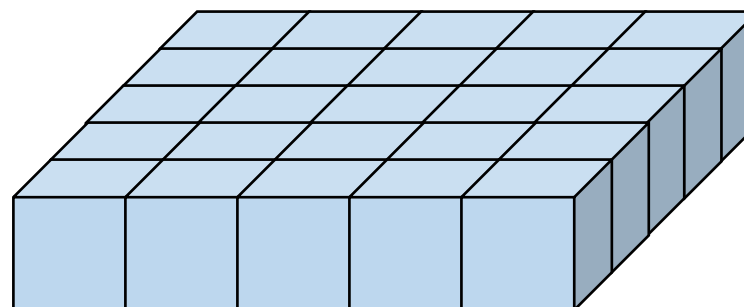


$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 3 \end{pmatrix}$$

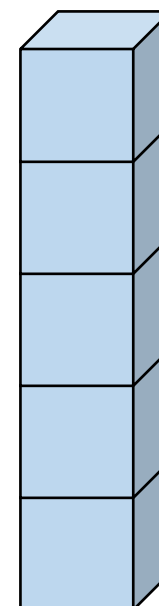


$$\log_5 125 = 3$$

=



×

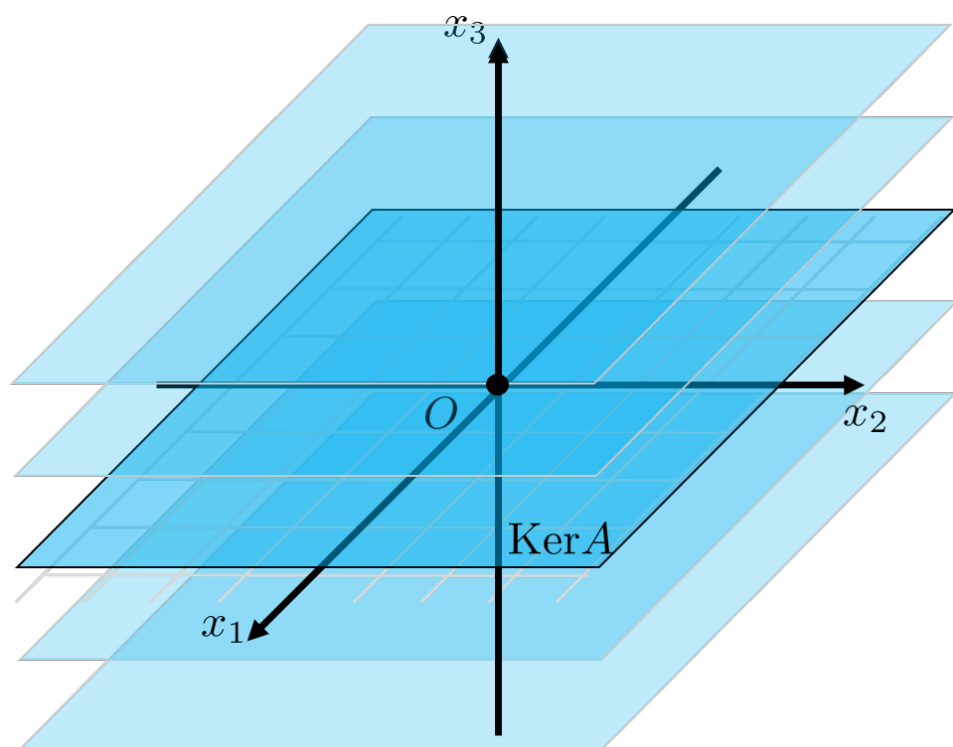


=

$$\log_5 25 = 2$$

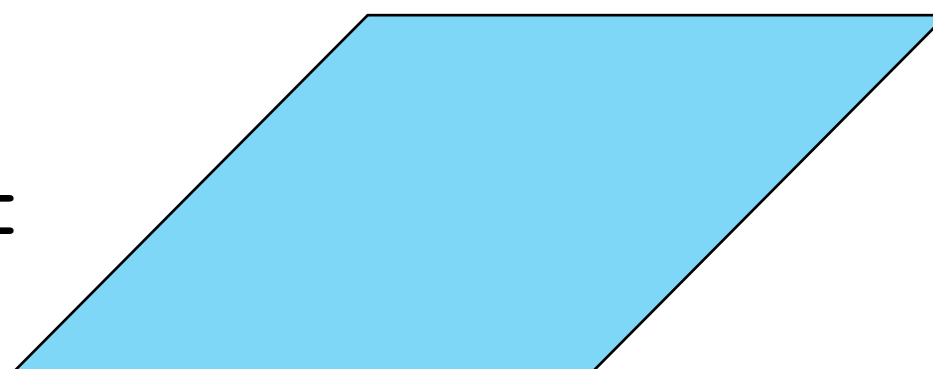
+

$$\log_5 5 = 1$$



$$\dim \mathbb{R}^3 = 3$$

=



×



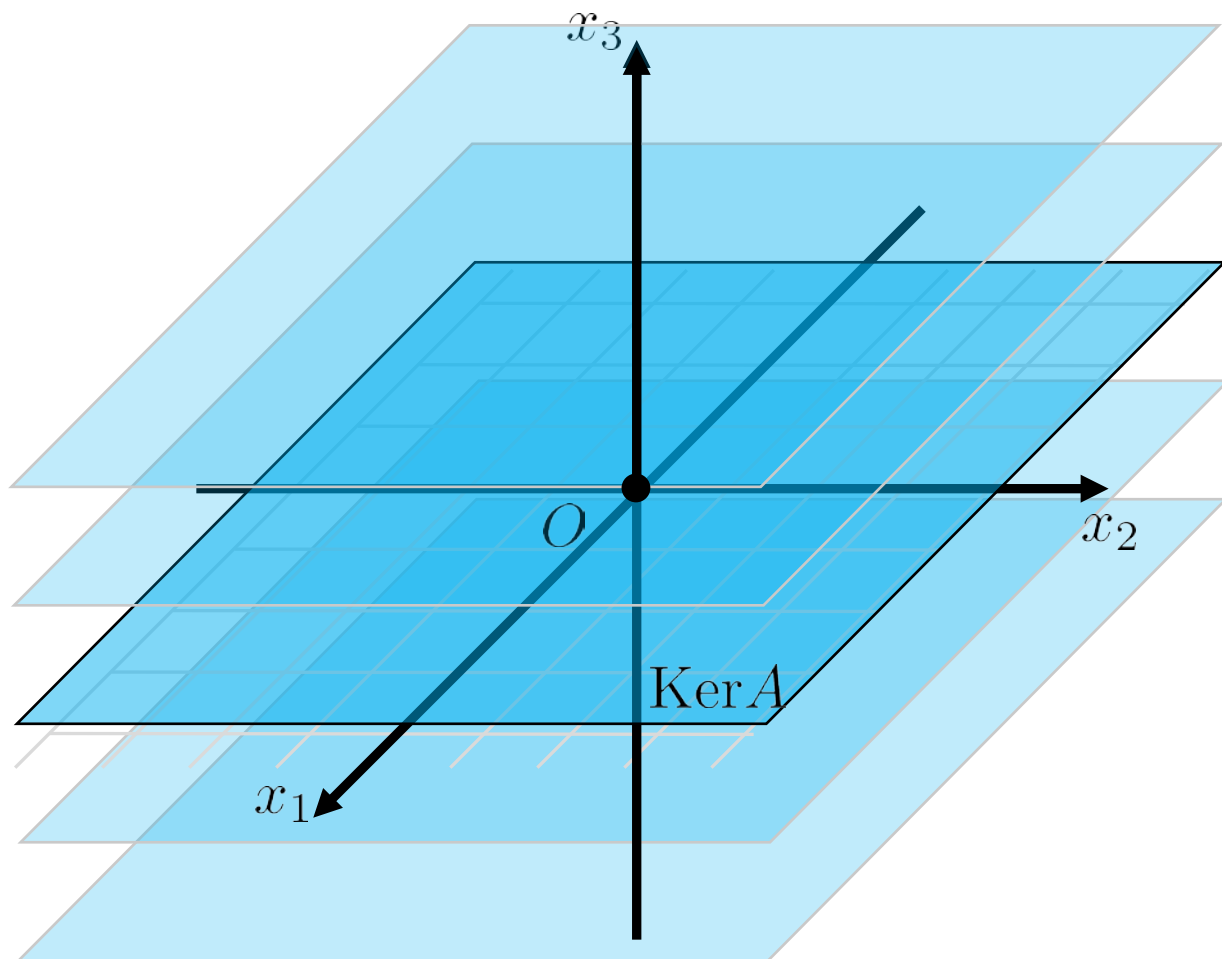
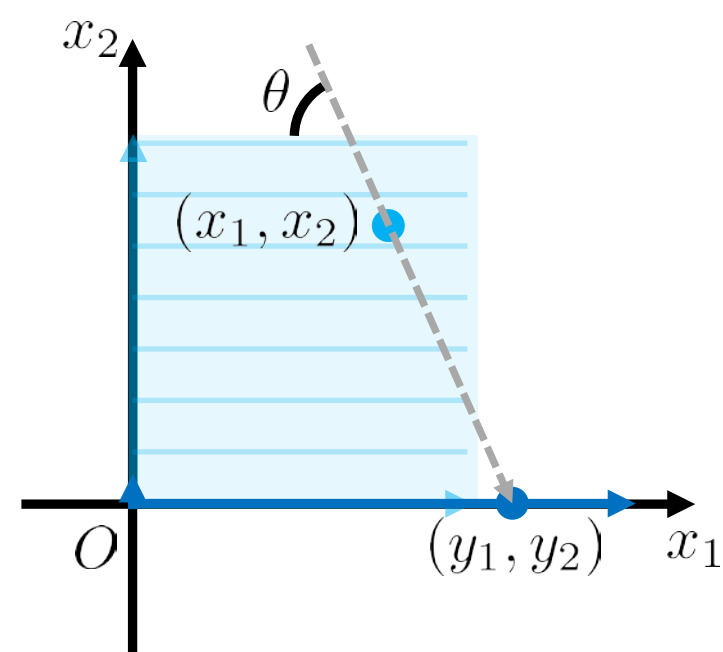
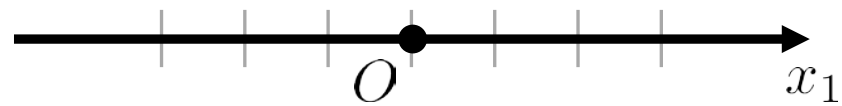
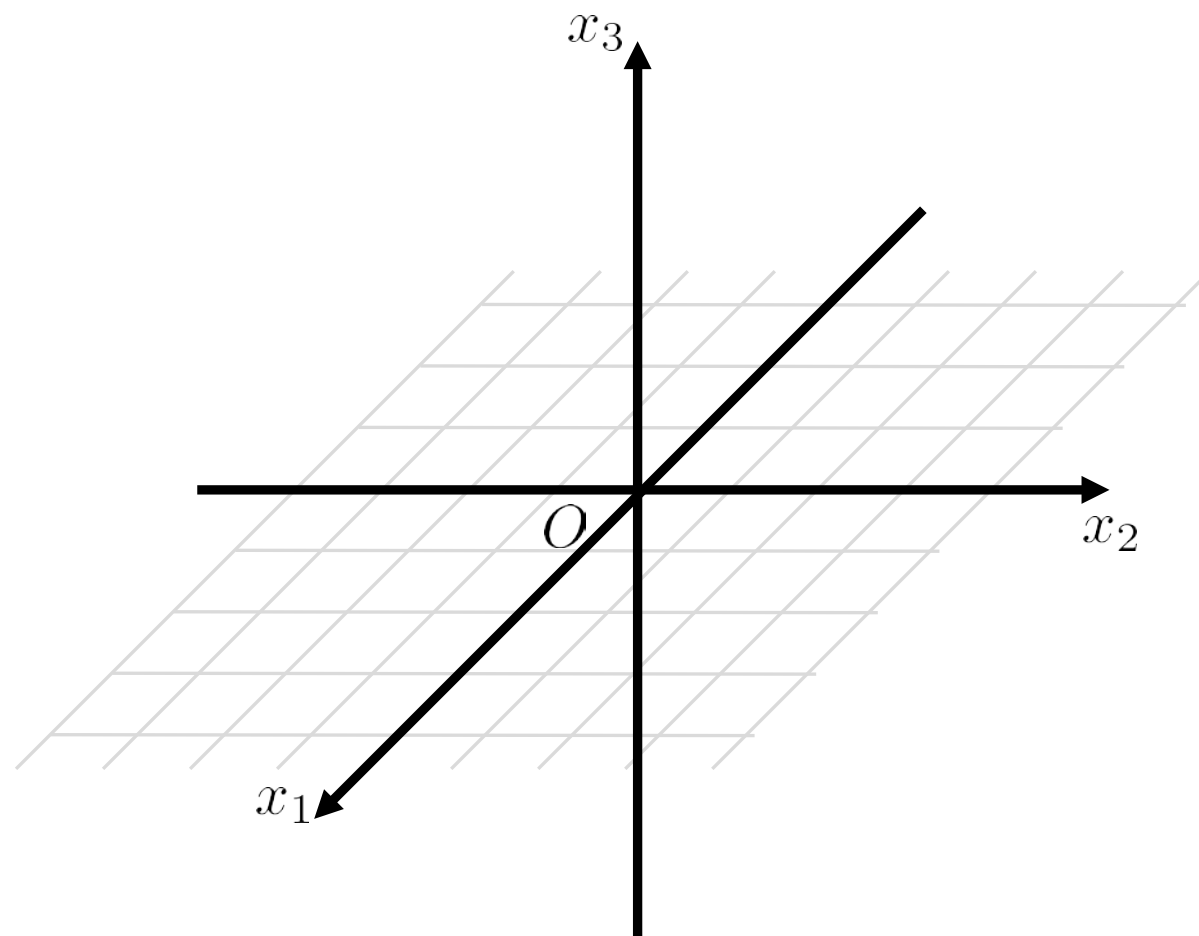
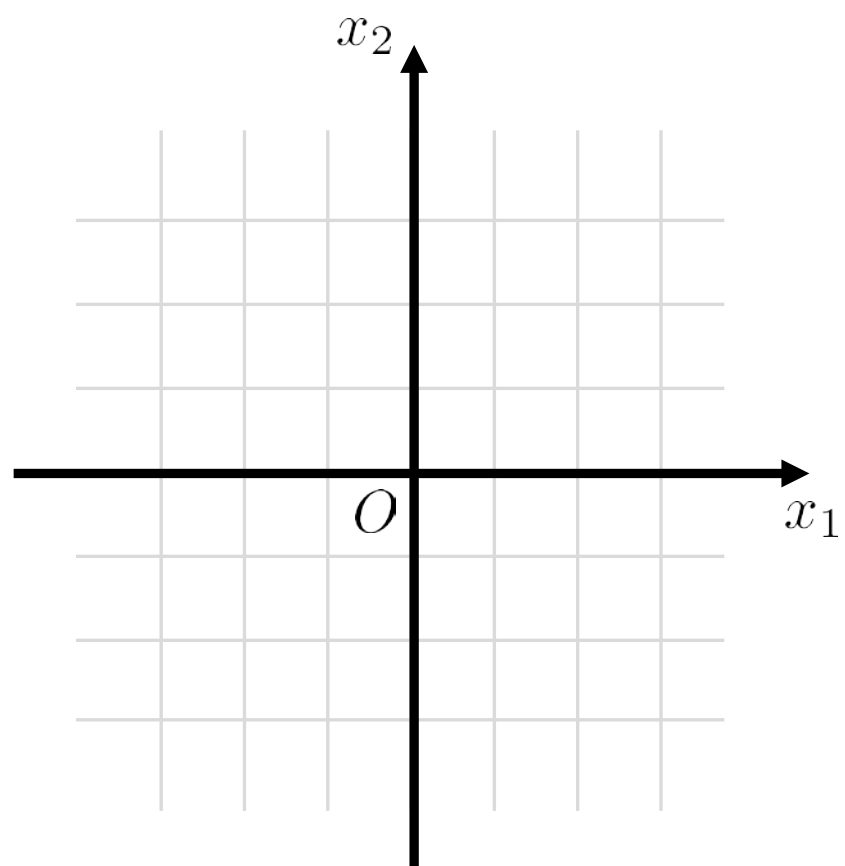
=

$$\dim \text{Ker} A = 2$$

+

$$\dim \text{Im} A = 1$$





$$(5) \quad \begin{cases} y_1 = 1 \cdot x_1 + \cot \theta \cdot x_2 \\ y_2 = 0 \cdot x_1 + 0 \cdot x_2 \end{cases}$$